
Revisiting ‘Little Jiffy, Mark IV’*

Towards a Bayesian KMO index

Emil Niclas Meyer-Hansen[†]

16 October 2025

Abstract

The *Kaiser-Meyer-Olkin* (KMO) index is a measure of sampling adequacy used by researchers to assess whether a data matrix is factorable prior to a factor analysis. Since its conception, the KMO index has remained a Frequentist statistic, leaving researchers unable to employ the advantages of Bayesian inference when assessing sampling adequacy. Building on the increasing relevance of the Bayesian statistical approach, as well as advancements in Markov-Chain Monte Carlo methods, the author proposes a re-conceptualization of the KMO index within the Bayesian framework that enables researchers to incorporate prior information and make coherent probabilistic statements about the sampling adequacy of a data matrix.©¹

Keywords: *Kaiser-Meyer-Olkin index, KMO, Bayesian Kaiser-Meyer-Olkin index, BKMO, Measure of Sampling Adequacy, MSA, Bayesian Measure of Sampling Adequacy, BMSA, Bootstrap KMO index, Robust KMO index, Robust BKMO index, Bayesian inference, Likelihood-based inference, Frequentist inference*

***Version:** 2025-10-16-16-01. **Citation:** Meyer-Hansen, E. N. (2025): ‘Revisiting ‘Little Jiffy, Mark IV’: Towards a Bayesian KMO index’, *Open Science Framework*, Working paper (v2025-10-16-16-01). DOI: 10.17605/OSF.IO/T3UPD

[†]**Author:** Independent Researcher, Aarhus, Denmark. Holds a Master of Science (MSc.) and Bachelor of Science (BSc.) in Political Science from Aarhus University, Denmark (email: emil098meyerhansen@gmail.com).

¹**License:** Except where otherwise indicated, all contents of this document and associated files are licensed under the *Creative Commons Attribution 4.0 International License* (CC BY 4.0). All software, including but *not* necessarily limited to, source code, executable code, code snippets, code chunks, algorithms, and/or scripts, attributable to this document and/or its associated files are expressly excluded from the foregoing license, and unless otherwise indicated, are instead licensed under the *GNU General Public License, version 3* (GPL-3.0). By engaging with this document and/or any associated files, which include, but are *not* necessarily limited to, downloading, using, viewing, and/or distributing any of them, in parts of whole, you agree to comply with the applicable license terms for the respective content types.

1 Introduction

The *Kaiser-Meyer-Olkin* (KMO) index, formally referred to as a *Measure of Sampling Adequacy* (MSA, Kaiser, 1970; Kaiser and Rice, 1974; Kaiser, 1981), is a statistic useful for determining whether a data matrix can be considered 'appropriate' for factor analysis (Dziuban and Shirkey, 1974; Field, 2018: 798; Revelle, 2025). Named in honor of Henry F. Kaiser², Edward P. Meyer³, and Ingram Olkin⁴ (cf. Dziuban and Shirkey, 1974; Kaiser, 1970: 405; Kaiser and Rice, 1974: 112; Kaiser, 1981: 380), it is often used to assess whether a data matrix is *factorable* (cf. Revelle, 2025, e.g., Field, 2018: 797-799), which is a desirable property to ascertain prior to an *exploratory factor analysis* (EFA, cf. Field, 2018: 778-833). While originally introduced as 'a second generation Little Jiffy'⁵ by Kaiser (1970), the currently most used iteration of the KMO index is the result of two subsequent revisions⁶, including a collaboration between Kaiser and John Rice⁷, titled 'Little Jiffy, Mark IV' (Kaiser and Rice, 1974).⁸ Though it historically may have seen most use within psychology (e.g., Dziuban and Shirkey, 1974; Zabihi et al., 2014; Ghadiri et al., 2015), often being used in conjunction with Maurice S. Bartlett's (1950) *test of sphericity* (e.g., Dziuban and Shirkey, 1974; Field, 2018), to this day, it remains a standard measure in statistical software (e.g., IBM, 2025; Revelle, 2025) that helps researchers across disciplines gauge whether their data can be meaningfully explored using factor analysis (e.g., Ahmad et al., 2024; Almeida et al., 2022; Sarfaraz et al., 2024; Sadowska et al., 2021; Danaei et al., 2024).

Conceived within the *classical statistical framework* (for introductory texts, see, e.g., Agresti, 2018; Field, 2018; Imai, 2017; Stock and Watson, 2019; Wooldridge, 2019), the KMO index can be shown to be conceptually 'frequentist' (cf. Kaiser, 1970; Kaiser and Rice, 1974; Kaiser, 1981), and a counterpart suitable for the alternative *Bayesian statistical framework* (for introductory texts, see, e.g., Gelman et al., 2014, 2021; Kruschke, 2014; Levy and Mislevy, 2020; McElreath, 2019) is currently missing. This is a relevant shortcoming, because these frameworks fundamentally differ in their conceptualization of probability and statistical uncertainty (see, e.g., Clayton, 2021; Field, 2018: 122-131; Imai, 2017: 242-244; Kruschke, 2014; Levy and Mislevy, 2020; McElreath, 2019). Given that recent advances in Bayesian inference (for a review, see, e.g., Gelman, 2022) have made it more viable for applied research (e.g., Gelman et al., 1996, 2019; Hoffman and Gelman, 2014; Vehtari et al., 2021) and the numerous emerging criticisms of Frequentism (e.g., Clayton, 2021; Cohen, 1994; Gelman, 2023; Gelman and Stern, 2006; Gigerenzer, 2004, 2018), transitioning to the Bayesian framework may be the best option for researchers (cf. Kruschke, 2014; McElreath, 2019; Wagenmakers et al., 2010). Accordingly, a multitude of frequentist statistics

²Former Professor of the University of California, Berkeley.

³Former professor of Loyola University, Chicago. It should be noted that there is some uncertainty about Edward P. Meyer being the 'Meyer' of 'Kaiser-Meyer-Olkin', because Henry F. Kaiser never appears to mention this 'Meyer' by first name, but only makes vague mentions of contributions from a 'Professor Meyer of Loyola, Chicago' (Kaiser, 1970: 405). At the same time, it seems that Kaiser did *not* himself coin the term 'Kaiser-Meyer-Olkin' as a name for the MSA, which instead appears to be an invention of Dziuban and Shirkey (1974), but these authors also make no explicit mention of a 'Professor Meyer'. However, based on personal communication with staff of Loyola University, Chicago, this 'Professor Meyer of Loyola, Chicago' is likely to be Edward Paul Meyer, who was a Professor of Psychology at Loyola during the period (1969 - 1972) when the KMO index was invented.

⁴Former Professor of Stanford University.

⁵The term 'Little Jiffy' was coined by former Professor Chester W. Harris of the University of Wisconsin-Madison and was used by Henry F. Kaiser to denote the analytic procedure of 'principal components with associated eigenvalues greater than one followed by normal varimax rotation' (cf. Kaiser, 1970: 402). Note that this 'Little Jiffy, Mark I' is *not* directly related to later versions of 'Little Jiffy' (e.g., Kaiser, 1970).

⁶The first revision of the KMO index, 'Little Jiffy, Mark III', went unpublished though its changes were incorporated into 'Mark IV' (cf. Kaiser and Rice, 1974: 112).

⁷Emeritus Professor of the University of California, Berkeley. Former Professor of the University of California, San Diego.

⁸A final revision of the KMO index was made by Kaiser (1981), which is simply the square-root of the original KMO index by Kaiser (1970), but this revised version appears mostly ignored across statistical software packages and applied research. Accordingly, this paper focuses on 'Little Jiffy, Mark IV' (Kaiser and Rice, 1974).

have been converted into Bayesian counterparts (e.g., Ben-Shachar et al., 2020; Gelman et al., 2019; Makowski et al., 2019a), but researchers acquainted with the KMO index may take issue with the lack of a Bayesian version that can properly exploit the numerous advantages of Bayesian inference, such as the ability to incorporate prior information into the analysis and make coherent probabilistic statements from a posterior distribution (cf. Clayton, 2021; Gelman et al., 2014; Kruschke, 2014; McElreath, 2019).

This paper solves the issue of a missing Bayesian KMO index by re-conceptualizing it within the Bayesian framework and demonstrating its implementation using standard statistical software. To build towards a Bayesian concept, this is achieved by revisiting the current KMO index (Section 2), discussing its mathematical and conceptual properties within the frequentist, likelihood-based, and Bayesian frameworks, as well as re-examining its viability for inference and robustness. The resulting *Bayesian Kaiser-Meyer-Olkin* (BKMO) *index* is then introduced and discussed (Section 3), being a *Bayesian Measure of Sampling Adequacy* (BMSA) that can incorporate prior information and express uncertainty about the 'sampling adequacy' (i.e., factorability) of a data matrix in the form of posterior distributions. With weakly informative priors and small samples, the BKMO index is effectively estimated using regularization, and with 'non-informative' priors, the BKMO index approximates results derived from its frequentist and likelihood-based counterparts. Analogous to its classical counterpart, the BKMO index can be made robust using rank-standardization, though a more Bayesian approach to robustness can also be achieved by changing the likelihood involved in its computation from a Gaussian to a Student t probability distribution. Similar to its bootstrapped frequentist version, the Bayesian version enables inferential statements to be made regarding the 'sampling adequacy' of the data matrix. The paper concludes by discussing the developed BKMO index (Section 4), reviewing its properties, areas of application, and robustness.

2 The Kaiser-Meyer-Olkin index

To build towards a Bayesian conceptualization, it is necessary to first understand the classical and most widely used version of the KMO index, that is, 'Little Jiffy, Mark IV' (Kaiser and Rice, 1974). This includes revisiting its mathematical formula, interpretation, and usability. Readers should note that this paper uses the terms 'KMO index', 'sampling adequacy', and 'factorability' interchangeably, since the KMO index *is* a measure of the factorability of a data matrix (cf. Kaiser, 1970; Kaiser and Rice, 1974; Kaiser, 1981). That notion also helps emphasize that the viability of the KMO index in applied research is dependent on the characteristics of the *data matrix* from which it is calculated. While explicitly specifying this data matrix is often neglected (e.g., Kaiser, 1970; Kaiser and Rice, 1974; Kaiser, 1981), this will be done here (Section 2.1), which serves to provide important information when reviewing the formula (Section 2.2) and interpretation (Section 2.3) of the KMO index. After showing its application (Section 2.4), its viability for inference is demonstrated (Section 2.5), followed by a discussion of its assumptions and robustness (Section 2.6).

2.1 Data matrix

Specifying the data matrix can be done using *set theory* (Burgess, 1948; Cantor, 1874) and standard notation (e.g., Levy and Mislevy, 2020) that is made more mnemonic for the purposes of the paper (cf. Kruschke, 2014). Suppose the existence of an n times k data matrix (i.e., $\mathcal{D}$). Let n be a positive integer denoting the total number of independent unit observations constituting a representative and sufficiently large random sample, taken from a

large and heterogenous population of units. This population could, in principle, be infinitely large, but for practical purposes, it is taken to be of finite size (i.e., N , with $\{n, N\} \subset \mathbb{N}$, $1 \ll n \leq N < \infty$), though at no loss of generality, with u indexing an arbitrary individual unit observation. Similarly, let k be a positive integer reflecting the total number of independently measured phenomena — theorized to be a set of manifestations (i.e., θ) of some *latent phenomenon* (i.e., ϕ) — that constitute a representative and sufficiently large random sample, taken from a large and heterogenous population of manifestations (i.e., Θ , with $\theta \subseteq \Theta$), with an arbitrary manifestation indexed by m . For similar reasons, the size of that population is also taken to be finite (i.e., K , with $\{k, K\} \subset \mathbb{N}$, and $1 \ll k \leq K < \infty$). This data matrix ($\mathcal{D}$) is illustrated in definition (2.1.1).

Assume that the latent phenomenon (ϕ) and all of its manifestations (θ) can be meaningfully expressed as *real-valued vectors* ($\{\phi, \theta\} \subset \mathbb{R}$, Boyd and Vandenberghe, 2018: 4; Stock and Watson, 2019: 748-749), and that the latent phenomenon is *independent and identically distributed* (*iid*, cf. McElreath, 2019: 81; Stock and Watson, 2019: 81-82) in the population following a *normal* probability distribution ($\mathcal{N}$, i.e., *Gaussian*, Gauss, 1809; cf. Agresti, 2018: 84-91; Imai, 2017: 286-292; Stock and Watson, 2019: 710-711; Wooldridge, 2019: 704-705) so that their values in the sample can be characterized as: $\phi_u \sim \mathcal{N}(\gamma, \iota^2)$, for u in $1, \dots, n$ (cf. Levy and Mislevy, 2020: 88), where γ is a real-valued *scalar* (Boyd and Vandenberghe, 2018: 4; Stock and Watson, 2019: 749) that denotes the mean parameter, while ι is a positive real-valued scalar denoting the standard deviation parameter.

$$\mathcal{D} \equiv \begin{bmatrix} \theta_{1,1} & \cdots & \theta_{1,k} \\ \vdots & \ddots & \vdots \\ \theta_{n,1} & \cdots & \theta_{n,k} \end{bmatrix} \quad (2.1.1)$$

Further, assume that the latent phenomenon is linearly distinct from its manifestations, with all manifestations also being linearly distinct from each other, meaning that ϕ and θ could together form a *full column rank matrix* (Stock and Watson, 2019: 751). Assume that the latent phenomenon temporally precedes in origin all of its manifestations (i.e., $\forall \theta_m : \phi \prec \theta_m$), and that every manifestation is temporally simultaneous in origin with every other manifestation (i.e., $\forall \theta_m : \theta_m \not\prec \theta_{m'}$), with m' denoting an arbitrary manifestation that it *not* m . Suppose that the latent phenomenon is *exogenous* of all other *causes* of the manifestations (i.e. ζ , so that $\forall \zeta : \phi \perp \zeta$).

In line with its standard conceptualization in factor analysis (Field, 2018: 781-785; see also Levy and Mislevy, 2020: 187-190), assume finally that the manifestations are *endogenous* and *iid* as a linear function of the latent phenomenon in the population, so that their values in the sample can be characterized with the following conditionally normal probability distribution: $\theta_{u,m} | \tau_m, \lambda_m, \varsigma_m^2 \sim \mathcal{N}(\tau_m + \lambda_m \phi_u, \varsigma_m^2)$, for u in $1, \dots, n$, and m in $1, \dots, k$. While ς_m is a positive and real k -length vector denoting the standard deviations for the manifestations, the mean vector parameter is decomposed into τ_m and λ_m , where the former is a real k -length vector denoting the intercepts, while the latter is a real k -length vector denoting the *factor loadings* (Field, 2018: 781; Levy and Mislevy, 2020: 144-145). These specifications and assumptions are discussed in section 2.6.

2.2 Formula

With this general data matrix ($\mathcal{D}$) just outlined in section 2.1 in mind, the steps typically used to calculate the KMO index can be shown (cf. Kaiser, 1981: 380; Kaiser and Rice, 1974; consistent with Dziuban and Shirkey, 1974: 359; IBM, 2025; Revelle, 2025). These steps include calculating the mean, variance, standard deviation,

covariance, correlation, and anti-image correlation across the manifestations in the data matrix. Conceptual considerations from the frequentist, likelihood-based, and Bayesian frameworks will be discussed throughout to build towards a Bayesian reconceptualization and show that the classical KMO index is a frequentist statistic.

2.2.1 Arithmetic mean

First, let $\text{Mean}(\theta_m)$ denote a *piecewise defined function* (cf. Stewart, 2010: 16-17) for calculating the arithmetic mean of manifestation m (cf. Agresti, 2018: 47-48; Field, 2018: 26-27; Gelman et al., 2021: 41-42; Stock and Watson, 2019: 82-83; Wooldridge, 2019: 666-668). Illustrated in definition (2.2.1), Σ denotes the *summation operation* (Cajori, 2007; Euler, 1755; Field, 2018: xxviii; Stock and Watson, 2019: 60), and the aforementioned u indexes the rows of $\mathcal{D}$ (i.e., unit observations), with n rows in total, while m indexes columns of $\mathcal{D}$ (i.e., manifestations). Here, the simplicity of function (2.2.1) helps express some of the conceptual differences between the frequentist, likelihood-based, and Bayesian frameworks, since the formula used in each are mathematically equivalent.

$$\text{Mean}(\theta_m) \equiv \begin{cases} \hat{\mu}_m \equiv \frac{1}{n} \sum_{u=1}^n \theta_{u,m} & \text{if Freq. } \wedge n < N & (2.2.1.1) \\ \mu_m \equiv \frac{1}{n} \sum_{u=1}^n \theta_{u,m} & \text{if Freq. } \wedge n = N & (2.2.1.2) \\ \bar{\theta}_m \equiv \frac{1}{n} \sum_{u=1}^n \theta_{u,m} & \text{if MLE-Bayes.} & (2.2.1.3) \end{cases} \quad (2.2.1)$$

While all these frameworks employ formula that equivalently calculates the equally-weighted arithmetic mean of manifestation m , the calculated mean can be conceptualized in at least three different ways. For example, a frequentist researcher is conceptually interested in estimating the *mean of the population* (μ_m). To achieve this, they conceptualize the formula used to arrive at μ_m as an *estimator* (cf. Stock and Watson, 2019: 105). The particular frequentist estimator of the population mean employed by formula (2.2.1.1) in function (2.2.1) can be denoted $\hat{\mu}_{\text{OLS}}$ to reflect its association with the estimation method of *ordinary least squares* (OLS, Legendre, 1805; Wooldridge, 2019: 70-73). Here, it is chosen *not* necessarily because it calculates the *mean of the sample* (i.e., $\bar{\theta}_m$), but because of its properties across an (in principle) infinitely large number of samples, where it can be considered an *unbiased estimator* of the population mean when relying on OLS, because it is equivalent to the expected sampling mean (i.e., $\mathbb{E}(\hat{\mu}_{\text{OLS}[m]}) = \mu_m$, Agresti, 2018: 116; Imai, 2017: 316-317; Stock and Watson, 2019: 106-108). Accordingly, formula (2.2.1.1) calculates the *estimated mean of the population* ($\hat{\mu}_m$). For frequentist researchers in possession of data on the entire population (i.e., $n = N$), the calculated mean is *conceptually* different from $\hat{\mu}_m$, since the sample mean here is the *population mean* (i.e., μ_m), enabling the use of formula (2.2.1.2). However, both of these conceptualizations are different from the likelihood-based and Bayesian approaches.

By comparison, formula (2.2.1.3) in function (2.2.1) can be considered *conceptually* Bayesian, because it calculates the *mean of the sample* (i.e., $\bar{\theta}_m$). This is because in the event of *no prior information*, formula (2.2.1.3) provides the best possible estimate of the mean of manifestation m given the data. Here, the Bayesian framework is *not* inherently built around estimating population characteristics through some concept of drawing infinite samples from that population (cf. Clayton, 2021; McElreath, 2019), and accordingly, there is no need for varying formula to distinguish between data from a sample or population. While (2.2.1.3) is here referred to as 'Bayesian', readers should note that this formula is only a *point estimator* (cf. Agresti, 2018: 115-116; Casella and Berger, 2002: 311) of the sample mean and fails to incorporate prior information. This leaves the calculated mean of formula

(2.2.1.3) conceptually equivalent to the *mode of the likelihood-distribution*, which is of interest to the *likelihood-based statistical framework* (cf. Agresti, 2018: 138-140; King, 1998). In this approach, the conceptual output of formula (2.2.1.3) is also preferred for function (2.2.1), since in a manner similar to the frequentist framework, it can be considered a *maximum-likelihood estimator* ($\hat{\mu}_{MLE}$), which is unbiased for the likelihood-based estimation method. This means that the conceptual output of formula (2.2.1.3) could alternatively be denoted the *maximum-likelihood estimate* (Agresti, 2018: 139; Stock and Watson, 2019: 405-406; Wooldridge, 2019: 725-726). As such, formula (2.2.1.3) is only conceptually Bayesian in context of no prior information, and it is otherwise mostly of interest to researchers subscribing to the likelihood-based framework.

It should be noted that while the likelihood and Bayesian approaches are conceptually distinct (cf. King, 1998: 21-30), for simplicity, when in the context of point estimators and no prior information, this paper makes no distinction between them and jointly refers to them as 'MLE-Bayesian'. For 'proper' Bayesian inference, which is concerned with the incorporation of prior information and the estimation of the entire posterior distribution, more complex calculations are required, which typically involve *Markov-Chain Monte Carlo* methods (MCMC, Brooks et al., 2011). This will be expanded upon when reconceptualizing the KMO index within the Bayesian framework (see Section 3.1).

Given the frequentist formula otherwise generally employed when calculating the KMO index (more on that in this section), the conceptual mean involved in the calculation of the classical KMO index (Kaiser and Rice, 1974) is taken to be the *estimated mean of the population* ($\hat{\mu}_m$), with the frequentist formula (2.2.1.1) thus used for function (2.2.1).

2.2.2 Variance

Second, let $\text{Var}(\theta_m)$ denote the piecewise function for calculating the *variance* of manifestation m (cf. Agresti, 2018: 54-55; Field, 2018: 28-31; Gelman et al., 2021: 41-42; Imai, 2017: 66-68; Stock and Watson, 2019: 83-84; Wooldridge, 2019: 695-696). As shown in function (2.2.2), *not* just the conceptual differences between the frequentist and MLE-Bayesian approaches but also their mathematical differences become apparent, since the three different conceptualizations of variance involve at least two mathematically distinct formula. These are similar because they all involve the calculation of the *sum of squares* (SS, cf. Agresti, 2018: 55; Field, 2018: 29, 56-57) of manifestation m , and the use of piecewise functions allows it to be conceptualized irrespective of statistical framework as: $SS \equiv \sum_{u=1}^n (\theta_{u,m} - \text{Mean}(\theta_m))^2$. What makes the function 'frequentist' or 'MLE-Bayesian' is the conceptual output of $\text{Mean}(\theta_m)$ and the weighing of the contributions of the observations. For example, formula (2.2.2.1) and (2.2.2.2) are frequentist, while (2.2.2.3) can be considered MLE-Bayesian.

$$\text{Var}(\theta_m) \equiv \begin{cases} \hat{\sigma}_m^2 \equiv \frac{1}{n-1} \sum_{u=1}^n (\theta_{u,m} - \hat{\mu}_m)^2 & \text{if Freq.} \wedge n < N & (2.2.2.1) \\ \sigma_m^2 \equiv \frac{1}{n} \sum_{u=1}^n (\theta_{u,m} - \mu_m)^2 & \text{if Freq.} \wedge n = N & (2.2.2.2) \\ s_m^2 \equiv \frac{1}{n} \sum_{u=1}^n (\theta_{u,m} - \bar{\theta}_m)^2 & \text{if MLE-Bayes.} & (2.2.2.3) \end{cases} \quad (2.2.2)$$

Formula (2.2.2.1) in function (2.2.2) is again a frequentist estimator (denoted $\hat{\sigma}_{OLS}^2$) that calculates the variance by dividing the SS with $n - 1$, which is a weight also known as *Bessel's correction* (cf. Field, 2018: 29-30; Stock and Watson, 2019: 112). This is a frequentist adjustment to the sample size based on the *degrees of freedom* that makes the estimator unbiased for OLS by accounting for the mean being estimated (i.e., $\mathbb{E}(\hat{\sigma}_{OLS[m]}^2) = \sigma_m^2$, cf. Field, 2018: 58-59; Stock and Watson, 2019: 112). This correction, however, is only applied when working

with samples (i.e., $n < N$), and frequentist researchers with census data (i.e., $n = N$) need *not* apply Bessel's correction but can instead rely on formula (2.2.2.2), since the population mean (μ_m) is known for such data (cf. Agresti, 2018: 56; Stock and Watson, 2019: 112).

For conceptual clarity, it should be noted that frequentist researchers may sometimes refer to $\hat{\sigma}_m^2$ as the 'sample variance' (denoting it s_m^2 , e.g., Field, 2018: 30; Gelman et al., 2021: 168; Imai, 2017: 325; Stock and Watson, 2019: 112), but this is *inappropriate* (cf. McElreath, 2019: 197), since $\hat{\sigma}_m^2$ is *not* the *sample* variance (i.e., s_m^2) but instead an *estimate of the population variance* (i.e., $\hat{\sigma}_m^2$), and for any finite sample, s_m^2 does *not* necessarily equal $\hat{\sigma}_m^2$. Instead, the 'sample variance', that is, the *variance in the sample* (s_m^2), is distinct and calculated by formula (2.2.2.3) in function (2.2.2), which despite being mathematically equivalent to formula (2.2.2.2) is conceptually different for the aforementioned reasons and thus preferred by Bayesian researchers (cf. McElreath, 2019: 197).

In line with the likelihood-based framework, the formula of (2.2.2.2) and (2.2.2.3) in function (2.2.2) is the *maximum-likelihood estimator* ($\hat{\sigma}_{MLE}^2$, cf. Casella and Berger, 2002: 313), in contrast to the *unbiased estimator* for OLS (σ_{OLS}^2 , cf. Field, 2018: 29-30; Stock and Watson, 2019: 112) used in formula (2.2.2.1). While formula (2.2.2.3) is generally preferred by MLE-Bayesians (cf. McElreath, 2019: 197), a 'proper' Bayesian estimate of the variance will again typically involve MCMC methods to estimate the posterior distribution and incorporate prior information.

Consistent with existing implementations (Dziuban and Shirkey, 1974; IBM, 2025; Revelle, 2025), the formula generally used by the current KMO index (Kaiser and Rice, 1974) for function (2.2.2) is based on formula (2.2.2.2), which outputs the *estimated population variance* ($\hat{\sigma}_m^2$). As such, the KMO index (cf. Kaiser and Rice, 1974) can be considered to be conceptually frequentist and *not* equivalent to a Bayesian or likelihood-based conceptualization for finite data, which is a detriment to researchers working within those frameworks who seek to assess 'sampling adequacy' prior to a factor analysis. The fact that the classical KMO index is conceptually frequentist is further corroborated in the following sections, where the remaining formula used in its calculation are discussed.

2.2.3 Standard deviation

Let $SD(\theta_m)$ then denote the piecewise function for calculating the *standard deviation* of manifestation m (cf. Agresti, 2018: 54-55; Field, 2018: 28-31; Gelman et al., 2021: 41-42; Imai, 2017: 66-68; Stock and Watson, 2019: 84; Wooldridge, 2019: 696). Illustrated in definition (2.2.3), where the *radical* (i.e., $\sqrt{}$) denotes the *square-root operation* (Descartes, 1637; Field, 2018: xxviii; Rudolff, 1525). Similar to the variance, the frequentist and Bayesian approaches differ with respect to $SD(\theta_m)$ as shown by conceptual disagreement about the output of function (2.2.3). Formula (2.2.3.1) and (2.2.3.2) are again frequentist, with (2.2.3.1) being the result of relying on formula (2.2.2.1) for calculating the variance (function 2.2.2), while (2.2.3.2) relies on formula (2.2.2.2). Formula (2.2.3.1) is again an estimator (i.e., $\hat{\sigma}_{OLS}$) and thus calculates the *estimated standard deviation in the population* (i.e., $\hat{\sigma}_m$), while (2.2.3.2) calculates the *population standard deviation* (i.e., σ_m), with the former being used when in possession of the sample, while the latter is reserved for census data, where the population mean (μ_m) is known.

Similar to the conceptualization of variance, frequentists often denote the conceptual output of formula (2.2.3.1) in function (2.2.3) the 'sample standard deviation' (i.e., s_m , e.g., Agresti, 2018: 55; Field, 2018: 30), but this is again *inappropriate* (cf. McElreath, 2019: 197), because it provides an *estimate of the population stan-*

dard deviation (i.e., $\hat{\sigma}_m$) that does *not* equal the sample standard deviation for finite data (Imai, 2017: 66-68). frequentist researchers should also note that while formula (2.2.3.1) is a less biased estimator than (2.2.3.2) when estimating the population standard deviation with OLS (due to formula 2.2.2.1 applying Bessel's correction), it is *not* unbiased due to Jensen's (1906) inequality (i.e., $\mathbb{E}(\hat{\sigma}_{\text{OLS}[m]}) \neq \sigma_m$, cf. Bolch, 1968). Instead, one will have to apply more complicated corrections (cf. Park et al., 2022; Park and Wang, 2020, 2022), which are typically *not* done when calculating the KMO index (cf. IBM, 2025; Revelle, 2025), meaning it relies on a biased estimator. Given the present aim of developing a *Bayesian* KMO index, these frequentist corrections will *not* be covered here.

$$\text{SD}(\theta_m) \equiv \begin{cases} \hat{\sigma}_m \equiv \sqrt{\hat{\sigma}_m^2} & \text{if Freq. } \wedge n < N & (2.2.3.1) \\ \sigma_m \equiv \sqrt{\sigma_m^2} & \text{if Freq. } \wedge n = N & (2.2.3.2) \\ s_m \equiv \sqrt{s_m^2} & \text{if MLE-Bayes.} & (2.2.3.3) \end{cases} \quad (2.2.3)$$

The last formula, (2.2.3.3), in function (2.2.3) can be considered MLE-Bayesian in the same manner as formula (2.2.2.3) in function (2.2.2), with formula (2.2.3.3) calculating the *sample standard deviation* (i.e., s_m , alternatively denoted $\hat{\sigma}_{\text{MLE}}$). MCMC is again the preferred method to estimate the entire posterior distribution and incorporate prior information rather than merely relying on the point estimator of formula (2.2.3.3).

In line with its reliance on formula (2.2.2.1) for function (2.2.2), and consistent with common implementations (IBM, 2025; Revelle, 2025), the classical KMO index is calculated using formula (2.2.3.1) in function (2.2.3), meaning it uses a *biased* estimate of the population standard deviation.

2.2.4 Covariance

Third, let $\text{Cov}(\theta_m, \theta_{m'})$ denote the piecewise function for calculating the *covariance* between manifestation m and m' (Agresti, 2018: 92; Field, 2018: 336-338; Imai, 2017: 384; Wooldridge, 2019: 697-698). Defined in (2.2.4), it should come as no surprise that the conceptual output of this function differs across approaches, where frequentists rely on formula (2.2.4.1) and (2.2.4.2), while MLE-Bayesians would apply (2.2.4.3). Again, formula (2.2.4.1) applies Bessel's correction to create an unbiased *estimator of the covariance in the population* (i.e., $\hat{\sigma}_{\text{OLS}[\theta_m, \theta_{m'}]}$, with $\mathbb{E}(\hat{\sigma}_{\text{OLS}[\theta_m, \theta_{m'}]}) = \sigma_{\theta_m, m'}$, Field, 2018: 336-338). Conceptually speaking, frequentists applying formula (2.2.4.1) are thus interested in an *estimate of the population covariance* ($\hat{\sigma}_{\theta_m, m'}$). Bessel's correction is again only applied when in possession of a sample, and with population data, frequentists apply formula (2.2.4.2) to calculate the *population covariance* (i.e., $\sigma_{\theta_m, m'}$), since the mean of each manifestation is known.

$$\text{Cov}(\theta_m, \theta_{m'}) \equiv \quad (2.2.4)$$

$$\begin{cases} \hat{\sigma}_{\theta_m, m'} \equiv \frac{1}{n-1} \sum_{u=1}^n (\theta_{u,m} - \hat{\mu}_m)(\theta_{u,m'} - \hat{\mu}_{m'}) & \text{if Freq. } \wedge n < N & (2.2.4.1) \\ \sigma_{\theta_m, m'} \equiv \frac{1}{n} \sum_{u=1}^n (\theta_{u,m} - \mu_m)(\theta_{u,m'} - \mu_{m'}) & \text{if Freq. } \wedge n = N & (2.2.4.2) \\ s_{\theta_m, m'} \equiv \frac{1}{n} \sum_{u=1}^n (\theta_{u,m} - \bar{\theta}_m)(\theta_{u,m'} - \bar{\theta}_{m'}) & \text{if MLE-Bayes.} & (2.2.4.3) \end{cases}$$

By contrast, MLE-Bayesians are conceptually more interested in the *sample covariance* (i.e., $s_{\theta_m, m'}$) and so do *not* apply Bessel's correction. Researchers applying either formula (2.2.4.2) or (2.2.4.3) for function (2.2.4) can alternatively be said to rely on a maximum-likelihood estimator (i.e., $\hat{\sigma}_{\text{MLE}[\theta_m, \theta_{m'}]}$), though 'proper' Bayesians generally substitute formula (2.2.4.3) with MCMC methods for its aforementioned advantages.

Consistent with existing implementations (IBM, 2025; Revelle, 2025), the frequentist nature of the current

KMO index is thus further corroborated due to calculating an *estimate of the population variance* ($\hat{\sigma}_{\theta_{m,m'}}$) by relying on formula (2.2.4.1) for function (2.2.4).

2.2.5 Correlation

Let $\text{Cor}(\theta_m, \theta_{m'})$ then denote the piecewise function for calculating the *Pearson product-moment correlation coefficient* (PPCC, cf. Bravais, 1844; Pearson, 1895a; Stigler, 1989) between manifestation m and m' :

$$\text{Cor}(\theta_m, \theta_{m'}) \equiv \begin{cases} \hat{\rho}_{m,m'} \equiv \frac{\hat{\sigma}_{\theta_{m,m'}}}{\hat{\sigma}_m \hat{\sigma}_{m'}} & \text{if Freq.} \wedge n < N & (2.2.5.1) \\ \rho_{m,m'} \equiv \frac{\sigma_{\theta_{m,m'}}}{\sigma_m \sigma_{m'}} & \text{if Freq.} \wedge n = N & (2.2.5.2) \\ r_{m,m'} \equiv \frac{s_{\theta_{m,m'}}}{s_m s_{m'}} & \text{if MLE-Bayes.} & (2.2.5.3) \end{cases} \quad (2.2.5)$$

which when applied between the k manifestations in $\mathcal{D}$, results in correlations that can be collected in a *correlation matrix*. This *square matrix* (Stock and Watson, 2019: 749), whose rows are indexed by i and columns by j , is composed of k^2 correlations, one for each relationship between the manifestations (including each manifestation with itself). This results in its off-diagonal being *symmetric* (Stock and Watson, 2019: 749) and bounded between -1 and 1 (i.e., due to the Cauchy-Bunyakovsky-Schwarz inequality; Bunyakovsky, 1859; Cauchy, 1821; Schwarz, 1888; Wooldridge, 2019: 698-699), reflecting the *linear correlation* (Imai, 2017: 143; Stock and Watson, 2019: 157) between manifestations (i.e., inter-correlations). By comparison, its diagonal is composed of a k -length *ones vector* (Boyd and Vandenberghe, 2018: 5), reflecting the perfect linear association of each manifestation with itself (i.e., intra-correlation). Given the obvious conceptual differences across the statistical approaches, the correlation matrix will be denoted $\hat{P}$ if relying on frequentist estimators for a sample, P if the data is the population, and R if relying on definitions preferable to the MLE-Bayesian approach.

With this denotation in mind, the inter-correlations are calculated as defined in function (2.2.5) (cf. Agresti, 2018: 92; Field, 2018: 338-340; Imai, 2017: 101-105, 384; Stock and Watson, 2019: 71; Wooldridge, 2019: 698-699), where the statistical framework used determines whether function (2.2.5) results in $\hat{P}$, P , or R . Given the aforementioned conceptualizations, $\hat{\rho}$ is the *estimated linear correlation in the population*, ρ is the *linear correlation in the population*, and r is the *linear correlation in the sample*.

Researchers should similarly note that while applying Bessel's correction makes the estimator of the variance and covariance unbiased, the OLS-estimator of the population linear correlation in formula (2.2.5.1) (i.e., $\hat{\rho}_{\text{OLS}[m,m']}$) is *not* an unbiased estimator of the population correlation (i.e., $\mathbb{E}(\hat{\rho}_{\text{OLS}[m,m]}) \neq \rho_{m,m'}$), again because of Jensen's (1906) inequality. While corrections to the correlations (or versions thereof) have been proposed, for example, Kelley's (1935) ϵ , Hays' (1963) ω , or *regularized* estimation (Schäfer and Strimmer, 2005), the literature lacks consensus on whether any of these are properly unbiased (e.g., Albers and Lakens, 2018; Carroll and Nordholm, 1975; Mordkoff, 2019). Since the typical calculation of the KMO index does *not* make any adjustments besides applying Bessel's correction for the aforementioned functions, these will *not* be further covered here.

By comparison, this issue of 'unbiasedness' can be avoided with formula (2.2.5.2) in function (2.2.5), which calculates the *population linear correlation* (i.e., $\rho_{m,m'}$) when in possession of census data. A conceptual alternative is formula (2.2.5.3), preferable to MLE-Bayesians, which calculates the *linear correlation in the sample* ($r_{m,m'}$, alternatively conceptualized as $\hat{\rho}_{\text{MLE}[m,m']}$). Formula (2.2.5.3) can again be expanded to estimate the entire posterior distribution and include prior information with the use of MCMC methods.

Consistent with the use of frequentist estimators to calculate the covariance and standard deviation (Dziuban and Shirkey, 1974; IBM, 2025; Revelle, 2025), the KMO index is conceptually frequentist and relies on the calculation of the *estimated correlation matrix in the population* ($\hat{P}$), applying the *biased* formula (2.2.5.1) for function (2.2.5).

2.2.6 Anti-image correlation

Finally, let $\text{Cor}_q(\theta_m, \theta_{m'})$ denote the piecewise function for calculating the *anti-image product-moment correlation coefficient* between manifestation m and m' (Kaiser, 1970: 405; Kaiser and Rice, 1974: 113-114; Kaiser, 1981: 379). This is a linear coefficient that is adjusted for the linear relationships between all the manifestations and is closely related to the *partial linear correlation* (Agresti, 2018: 343-346, 362; Field, 2018: 358-361). Again, when applied to every manifestation in $\mathcal{D}$, the resulting correlations can be used to construct a k times k *anti-image correlation matrix* (consistent with Kaiser, 1970: 405; Kaiser and Rice, 1974: 113-114; Kaiser, 1981: 379), which, similar to the correlation matrix, is square and symmetric, with values ranging from -1 to 1, though its diagonal is instead a negative ones vector. The anti-image inter-correlations can be calculated with reference to the k rows and k columns of the correlation matrix (where $i \neq j$):

$$\text{Cor}_q(\theta_m, \theta_{m'}) \equiv \begin{cases} \hat{\rho}_{q[i,j]} \equiv -\frac{\hat{\rho}_{i,j}^{-1}}{\sqrt{\hat{\rho}_{i,i}^{-1}\hat{\rho}_{j,j}^{-1}}} & \text{if Freq.} \wedge n < N & (2.2.6.1) \\ \rho_{q[i,j]} \equiv -\frac{\rho_{i,j}^{-1}}{\sqrt{\rho_{i,i}^{-1}\rho_{j,j}^{-1}}} & \text{if Freq.} \wedge n = N & (2.2.6.2) \\ r_{q[i,j]} \equiv -\frac{r_{i,j}^{-1}}{\sqrt{r_{i,i}^{-1}r_{j,j}^{-1}}} & \text{if MLE-Bayes.} & (2.2.6.3) \end{cases} \quad (2.2.6)$$

The conceptual output of $\text{Cor}_q(\theta_m, \theta_{m'})$ again depends entirely on the used statistical framework and data, with $\hat{P}_q$ being the *estimated anti-image correlation matrix in the population*, whose elements are given by formula (2.2.6.1) in function (2.2.6), while P_q denotes the *anti-image correlation matrix in the population*, and formula (2.2.6.2) is thus used for census data. MLE-Bayesians can conceptualize the output of $\text{Cor}_q(\theta_m, \theta_{m'})$ as R_q to denote the *anti-image correlation matrix in the sample*, with elements given by formula (2.2.6.3).

In line with the previous considerations, it is apparent that this frequentist KMO index conceptually involves the *estimated anti-image correlation in the population* ($\hat{P}_q$) through its reliance on formula (2.2.6.1) for function (2.2.6) (cf. Dziuban and Shirkey, 1974; IBM, 2025; Revelle, 2025).

For all these conceptualizations, it should be noted that the calculation of $\text{Cor}_q(\theta_m, \theta_{m'})$ necessitates that the correlation matrix is *invertible* (Boyd and Vandenberghe, 2018: 202), because it relies on the calculation of the *inverse correlation matrix* (cf. Kaiser and Rice, 1974: 113), whose components are denoted as $\hat{\rho}^{-1}$, ρ^{-1} , and r^{-1} in function (2.2.6), respectively. This necessity is the reason why the aforementioned stipulations of the data matrix ($\mathcal{D}$) in section 2.1 explicitly specified that the manifest phenomena were all linearly distinct and could form a full column rank matrix, because any instance of *perfect multicollinearity* in the correlation matrix makes it *singular* and thus *not invertible* (Boyd and Vandenberghe, 2018: 202-203). Another requirement is that piecewise functions (2.2.1) - (2.2.6) all require complete data, meaning that no data can be missing. While Kaiser and Rice (1974: 113) recommend using *mean imputation* to handle missing values, an arguably more appropriate procedure would involve *multiple imputations by chained equations* (MICE, see van Buuren and Groothuis-Oudshoorn, 2011) to improve imputation accuracy and account for imperfect imputations.

2.2.7 KMO index

Now, using the values computed from these standard formula, let $\text{KMO}(\mathcal{D})$ denote the piecewise function for calculating the overall 'sampling adequacy' of a data matrix (Kaiser and Rice, 1974: 112-113; consistent with Dziuban and Shirkey, 1974; IBM, 2025; Revelle, 2025). Since it has become apparent that the functions involved in the calculation of the KMO index differ in conceptualization and mathematical formula, $\text{KMO}(\mathcal{D})$ can be made to account for these variations, as defined in function (2.2.7). Here, $\sum \sum$ denotes the row-wise summation operation, $\hat{\alpha}$ denotes the *estimated sampling adequacy in the population* (formula 2.2.7.1), α denotes the *sampling adequacy in the population* (formula 2.2.7.2), and a denotes the *sampling adequacy in the sample* (formula 2.2.7.3), which help distinguish the conceptual nuances of the KMO index. Calculating the individual KMO indices, that is, the MSA for manifestation m in the data ($\mathcal{D}$), is given by the similarly distinguishing piecewise function (2.2.8).

The reason for distinguishing between the varying conceptual outputs produced by the formula in function (2.2.7) and (2.2.8) should have been made apparent based on the discussion of the different concepts and formula involved in their calculation. While it will here be left to the discretion of the researcher to determine whether they are interested in the 'sampling adequacy in the *population*' (e.g., formula 2.2.7.1 - 2.2.7.2) or the 'sampling adequacy in the *sample*' (e.g., formula 2.2.7.3), the latter MLE-Bayesian conceptualization does appear more consistent with the aim of the KMO index (see Kaiser, 1970, 1981; Kaiser and Rice, 1974), since it provides a metric about the factorability of the data matrix that was collected, instead of some hypothetical metric about infinitely many sampled data matrices.

$$\text{KMO}(\mathcal{D}) \equiv \begin{cases} \hat{\alpha} \equiv \frac{\sum_{i \neq j} \hat{\rho}_{i,j}^2}{\sum_{i \neq j} \hat{\rho}_{i,j}^2 + \sum_{i \neq j} \hat{\rho}_{q[i,j]}^2} & \text{if Freq. } \wedge n < N & (2.2.7.1) \\ \alpha \equiv \frac{\sum_{i \neq j} \rho_{i,j}^2}{\sum_{i \neq j} \rho_{i,j}^2 + \sum_{i \neq j} \rho_{q[i,j]}^2} & \text{if Freq. } \wedge n = N & (2.2.7.2) \\ a \equiv \frac{\sum_{i \neq j} r_{i,j}^2}{\sum_{i \neq j} r_{i,j}^2 + \sum_{i \neq j} r_{q[i,j]}^2} & \text{if MLE-Bayes.} & (2.2.7.3) \end{cases} \quad (2.2.7)$$

$$\text{KMO}(\mathcal{D}_m) \equiv \begin{cases} \hat{\alpha}_m \equiv \frac{\sum_{j,j \neq i} \hat{\rho}_{i,j}^2}{\sum_{j,j \neq i} \hat{\rho}_{i,j}^2 + \sum_{j,j \neq i} \hat{\rho}_{q[i,j]}^2} & \text{if Freq. } \wedge n < N & (2.2.8.1) \\ \alpha_m \equiv \frac{\sum_{j,j \neq i} \rho_{i,j}^2}{\sum_{j,j \neq i} \rho_{i,j}^2 + \sum_{j,j \neq i} \rho_{q[i,j]}^2} & \text{if Freq. } \wedge n = N & (2.2.8.2) \\ a_m \equiv \frac{\sum_{j,j \neq i} r_{i,j}^2}{\sum_{j,j \neq i} r_{i,j}^2 + \sum_{j,j \neq i} r_{q[i,j]}^2} & \text{if MLE-Bayes.} & (2.2.8.3) \end{cases} \quad (2.2.8)$$

Accordingly, the classical KMO index has been shown to be a frequentist statistic and a Bayesian reconceptualization can be considered appropriate and in line with its intended purpose. While this section has demonstrated the calculation of the KMO index, it is equally important to know how to meaningfully interpret it.

2.3 Interpretation

Irrespective of conceptualization or formula, the 'sampling adequacy' measured by the KMO index can be meaningfully interpreted as the ratio of the sum of squared inter-correlations relative to the sum of the squared inter-correlations and squared anti-image inter-correlations. Given the specifications in section 2.1, this makes the KMO index a real scalar ($\text{KMO} \in \mathbb{R}$, for exceptions, see below), where a value of .5 occurs when the sum of squared inter-correlations are equal to the sum of the squared anti-image inter-correlations (cf. Kaiser and Rice, 1974: 113).

Regarding the theoretical lower and upper bounds of the KMO index, there have been made numerous conflicting claims. Early researchers assert that its bounds are 0 and 1 (Dziuban and Shirkey, 1974: 359; Kaiser and Rice, 1974: 113), where a value of 0 would indicate that 'the sum of correlations is large relative to the sum of correlations, indicating diffusion in the pattern of correlations' (Field, 2018: 798), while a value close to 1 would indicate that 'patterns of correlations are relatively compact and so factor analysis should yield distinct and reliable factors.' (Field, 2018: 798). These bounds were challenged by (Shirkey and Dzuban, 1976), who by simulating independent variables mostly failed to produce KMO index values below .5 (as cited in Kaiser, 1981: 380). By comparison, examinations of real-world data from Harman (1967), Armstrong and Soelberg (1968) yielded a KMO index as high as .93 (cf. Dziuban and Shirkey, 1974: 360) and as low as .42 (cf. Dziuban and Shirkey, 1974: 359). In an attempt to fix what was perceived as non-intuitive bounds, (Kaiser, 1981) abandoned the 'Mark IV' revision (i.e., Kaiser and Rice, 1974) in favor of the original 'Mark II' version (i.e., Kaiser, 1970), whose lower bound instead was negative (cf. Kaiser, 1981: 381). To increase interpretation, Kaiser (1981) revised 'Mark II' to instead be its square-root, but this inadvertently meant that the lower bound of this new 'Mark V' became an *imaginary number* (Kaiser, 1981: 381; cf. Nahin, 1998), and that version has seen little use compared to 'Mark IV' (i.e., Kaiser and Rice, 1974).⁹

To solve this continuing discrepancy and obtain more clarity about the theoretical bounds of the 'Mark IV' KMO index (i.e., Kaiser and Rice, 1974), which in turn will be informative about the bounds of its Bayesian reconceptualization, this paper makes some analytic observations and briefly investigates the sampling characteristics of the KMO index with a series of *Monte Carlo* simulations (Robert and Casella, 2013; Rubinstein and Kroese, 2016) using the R *programming language*¹⁰ (R Core Team, 2025). These were partly motivated by Kaiser's (1970: 405) statements about 'Mark II' (i.e., Kaiser, 1970) that, *ceteris paribus*, the value of the KMO index increases as a function of n , k , and $\text{Cor}(\theta_m, \theta_{m'})$, while it decreases as a function of $\text{Cor}_q(\theta_m, \theta_{m'})$. Following this, 480,060 simulations were conducted that directly manipulated the dependency structure of the correlation matrix, the magnitude and direction of the correlations, and the number of theorized manifestations (k), to gauge the association between the inter-correlations and anti-image inter-correlations, and the values of the KMO index under linearly interdependent and independent (i.e., random) correlation structures, expanding upon the simulation study by Shirkey and Dzuban (1976).

Before consulting the simulated results, the analytic observations made were these: It can be stated that for any correlation matrix composed of perfectly linearly independent variables (i.e., its off-diagonals are zero vectors) that the KMO index is *undefined* for the real number system, because both the inter-correlations and the anti-image inter-correlations become zero, resulting in a division by zero in the formula of functions (2.2.7) and (2.2.8); it can be further stated that for any correlation matrix with perfectly linearly dependent variables (i.e., its off-diagonals are ones vectors) that the KMO index is also undefined, because the correlation matrix is singular (as similarly observed by Kaiser and Rice, 1974: 114), making it impossible to calculate the inverse inter-correlations needed in function (2.2.6). This inadvertently means that correlation matrices that could intuitively constitute lower and upper bounds does *not* yield a KMO index of 0 or 1.

These insights are complimented by the simulations, which reveal that the association between the correla-

⁹For researchers interested in these mostly forgotten versions of the KMO index, their mathematical formula, and functions implementing them in R (R Core Team, 2025), are provided in section 6.3 of the Appendix.

¹⁰For software specifications, see Section 6.1 of the Appendix

tion matrix and anti-image correlation matrix is non-monotonous and complex, being independent, positive, and negative under different circumstances. This means that an increase in the inter-correlations does *not* necessarily lead to an improved KMO index because the anti-image inter-correlations may similarly increase. This complexity may be a cause of why data matrices with 'bad' factorability often yield a KMO index of .5. Regarding the theoretical bounds, the lowest calculated KMO index was 2.54×10^{-09} , while the highest was 9.98×10^{-01} (for more information, see Section 6.4 of the Appendix).

These results are consistent with the earliest claims that the theoretical bounds of the KMO index are indeed 0 and 1, though based on the analytic insights, its bounds are more likely exclusive rather than inclusive (i.e., $\text{KMO} \in \mathbb{R}^{(0;1)}$ rather than $\text{KMO} \in \mathbb{R}^{[0;1]}$). However, they also corroborate claims (e.g., Field, 2018: 798) that values below .5 and values close to 1 (e.g., $> .95$) are highly unlikely, occurring across these simulations at a frequency of 6.02% and 3.03%, respectively. Another key insight is that a KMO index of exactly .5 is a highly possible outcome, occurring across the simulations at a frequency of 83.2%. Since this value occurred consistently and almost irrespective of how 'good' or 'bad' the inter-correlations were, a KMO index value of .5 can be taken to provide indecisive information in an assessment of the 'sampling adequacy' of the data matrix, which will be considered later when discussing inferential viability (see Section 2.5). Taken together, and with the mentioned caveats in mind, higher values of the KMO index thus generally indicate a greater percentage of shared information between the theorized manifestations (i.e., factorability), which is desirable for a factor analysis.

Table 1. Interpretive guidelines wrongly associated with the KMO index

Threshold	Interpretation
$1 \geq \text{IFS} \geq .9$	'Marvelous'
$.9 > \text{IFS} \geq .8$	'Mertitourious'
$.8 > \text{IFS} \geq .7$	'Middling'
$.7 > \text{IFS} \geq .6$	'Mediocre'
$.6 > \text{IFS} \geq .5$	'Miserable'
$.5 > \text{IFS} \geq 0$	'Unacceptable'

NOTE: Interpretive guidelines developed for the *Index of Factor Simplicity* (IFS, Kaiser, 1974: 35; see also Kaiser and Rice, 1974: 112), which are often incorrectly associated with the KMO index (e.g., Dziuban and Shirkey, 1974: 359; Revelle, 2025).

To help researchers make discrete decisions of 'sampling adequacy', guidelines by Kaiser (1974), reproduced in table 1, are often referenced when interpreting the KMO index (e.g., Dziuban and Shirkey, 1974: 359; Revelle, 2025). While these guidelines may be appropriate for interpreting it, they were *not* originally formulated in relation to the KMO index. In the paper referenced as the source of these guidelines (i.e., Kaiser, 1974; cf. Dziuban and Shirkey, 1974: 359; Revelle, 2025), Kaiser explicitly introduces them for a separate *Index of Factor Simplicity* (IFS, Kaiser, 1974: 35; see also Kaiser and Rice, 1974: 112), and the only relationship between these guidelines and 'sampling adequacy' is that the IFS is partly based on the KMO index (see Kaiser and Rice, 1974). Instead, in reference to the original 'Mark II' version of the KMO index, Kaiser (1970: 405) states that 'good factor-analytic data' requires a KMO index of at least .8, while 'really excellent data' only occurs with a KMO index of at least .9. Whether these generalize to the 'Mark IV' covered here is *not* mentioned in Kaiser and Rice (1974), and Kaiser

(1981) fails to comment on Dziuban and Shirkey's (1974: 359) use of the IFS-guidelines for interpreting the KMO index. However, given that Kaiser acknowledges the substantial differences between 'Mark II' and 'Mark IV' (Kaiser, 1981: 380-381), it is unlikely that he would favor interpreting them using the same guidelines.

While it can be useful to derive interpretive heuristics, instead of developing them entirely on a theoretical basis and applying them across fields indiscriminately (e.g., Cohen, 1988), one could argue that they should instead be empirically based and specified to suit the particular contexts of each field (cf. Funder and Ozer, 2019; Gignac and Szodorai, 2016; Lovakov and Agadullina, 2021). The currently missing guidelines for interpreting the KMO index could thus be addressed through an empirical examination of KMO indices in the literature, with interpretive guidelines being specified for each particular field. At the same time, one could also argue that the discretization involved in producing such rules-of-thumb of a continuous measure likely oversimplifies interpretation (cf. Field, 2018: 117), and researchers should thus ideally use guidelines with caution and report non-discretized KMO-indices. It is left to the reader's discretion which of these perspectives to take, and this paper makes no attempt to empirically derive guidelines for interpreting the KMO index. To more practically illustrate the utility of the KMO index, an example using simulated data is provided in the subsequent section.

2.4 Example

To demonstrate the usefulness of the KMO index, a 1,000 times 10 data matrix is simulated, again with *Monte Carlo* methods (Rubinstein and Kroese, 2016) using R (R Core Team, 2025). In line with the aforementioned stipulations in section 2.1, the latent phenomenon (ϕ) is generated with the standard normal parameter specifications: $\gamma = 0$ and $\iota = 1$, while its manifestations (θ) are generated with the specifications: $\forall \theta_m : \tau_m = 0, \lambda_m \sim \mathcal{U}(.3, .7), \varsigma_m = 1$, with $\mathcal{U}$ being the continuous *uniform probability distribution* (cf. Gelman et al., 2014: 577-578; McElreath, 2019: 36), where .3 inclusively denotes the lower bound and .7 the upper bound. While the specification of 10 manifestations is rather arbitrary and merely serves to help demonstrate a computation of the KMO index, this number has previously been considered adequate for reliably measuring latent phenomena (e.g., Ashton and Lee, 2009). The *smallest effect size of interest* (SESOI, see Lakens, 2017) is specified to be .3, since, for simplicity, this constitutes a general threshold for the lowest factor loading of interest (cf. Field, 2018: 806). Similarly, the specified sample size of 1,000 is also arbitrary, though in relation to EFA, it has been interpreted as 'excellent' (Comrey and Lee, 1992; Field, 2018: 797), and it provides a 99.9% *statistical power* (Cohen, 1988, 1991) to detect a factor loading as low as .3, which can be considered more than 'adequate' (cf. Cohen, 1988, 1991). The upper threshold of a factor loading is specified to be .7 to mitigate perfect multicollinearity among the 10 manifestations (cf. Field, 2018: 799; McElreath, 2019), which would make the correlation matrix singular and the KMO index undefined. This issue can be assessed by computing the *variance inflation factor* (VIF, Agresti, 2018: 445-446; Lüdtke et al., 2021; Wooldridge, 2019: 92) when predicting one manifestation using all other manifestations, and from its results, the issues of multicollinearity can be considered 'low' (i.e., $\forall \theta_m : VIF \leq 1.091$, Ben-Shachar et al., 2020; Zuur et al., 2010).

To ease model specification (cf. McElreath, 2019: 111), all measures are transformed using *Fisher z-scoring* (henceforth *standardization*, cf. Field, 2018: 37; Stock and Watson, 2019: 65; Wooldridge, 2019: 696-697). In line with the ongoing conceptual distinction of formula between statistical frameworks, let $Z(\theta_m)$ denote the piecewise function for standardizing manifestation m , defined in (2.4.1), where formula (2.4.1.1) is a frequentist

standardization based on the *estimated mean* (i.e., $\hat{\mu}_m$) and (*biased*) *standard deviation* (i.e., $\hat{\sigma}_m$) of the population. Formula (2.4.1.2) is similarly frequentist and is used with census data, with standardization being based on the *mean* (i.e., μ_m) and *standard deviation* (i.e., σ_m) of the population. By contrast, formula (2.4.1.3) can be considered MLE-Bayesian, with the standardization being based on the *mean* (i.e., $\bar{\theta}_m$) and *standard deviation* (i.e., s_m) of the sample. This could again be made 'properly' Bayesian using MCMC methods, though that is beyond the scope of this paper. For conceptual clarity, this demonstration of the frequentist KMO index uses formula (2.4.1.1) for function (2.4.1), and applying this function across every column of the data matrix simplifies the computation of the correlation matrix, because it becomes equivalent to the variance-covariance matrix (i.e., $\text{Cov}(Z(\theta_m), Z(\theta_{m'})) = \text{Cor}(\theta_m, \theta_{m'})$).

$$Z(\theta_m) \equiv \begin{cases} \frac{\theta_{u,m} - \hat{\mu}_m}{\hat{\sigma}_m} & \text{if Freq. } \wedge n < N & (2.4.1.1) \\ \frac{\theta_{u,m} - \mu_m}{\sigma_m} & \text{if Freq. } \wedge n = N & (2.4.1.2) \\ \frac{\theta_{u,m} - \bar{\theta}_m}{s_m} & \text{if MLE-Bayes.} & (2.4.1.3) \end{cases} \quad (2.4.1)$$

To help illustrate the usefulness of standardization, it is important to recognize that numerous statistical functions can be conceptualized as special cases of the *generalized linear model* (GLM, Field, 2018; Gelman et al., 2021: 263-264; Kruschke, 2014), and the correlations of a correlation matrix are no exception. For reasons that will become apparent when demonstrating the Bayesian KMO index (Section 3.3), a GLM-expression for calculating the correlations between the manifestations in the data, using standard notation (e.g., Levy and Mislevy, 2020; McElreath, 2019: 441-442), is provided in model (2.4.2).

$$Z(\theta)_u | \mu_m, \Omega \sim \mathcal{MVN}(\mu_m, \Omega) \text{ for } u \text{ in } 1, \dots, n \quad (2.4.2)$$

$$\Omega = DPD$$

$$\mu_m = 0 \text{ for } m \text{ in } 1, \dots, k$$

$$\text{diag}(D)_m = 1 \text{ for } m \text{ in } 1, \dots, k$$

Here, each row (u) in the standardized set of k manifestations, with $Z(\theta)_u = \{Z(\theta_{u,1}), \dots, Z(\theta_{u,k})\}$, is taken to be *iid* as a conditionally *multivariate normal* probability distribution (i.e., $\mathcal{MVN}$, Gelman et al., 2014: 578, 582; Stock and Watson, 2019: 752-753), with a k -length manifestation-varying mean vector parameter (i.e., μ_m) and a k times k variance-covariance matrix parameter (i.e., Ω). Following Barnard and colleagues (2000; see also McElreath, 2019: 441-442), the variance-covariance matrix can be decomposed into two k times k *diagonal matrices* (Boyd and Vandenberghe, 2018: 114) of standard deviations (i.e., D) and one k times k correlation matrix (i.e., P). Due to the aforementioned standardization of the data, and the absence of predictors in the model, the mean vector becomes a k -length *zero vector* (Boyd and Vandenberghe, 2018: 5) of intercepts; and D becomes a k times k *identity matrix* (Boyd and Vandenberghe, 2018: 113; Stock and Watson, 2019: 749), because its diagonal becomes a ones vector, which simplifies the variance-covariance matrix (Ω) into the correlation matrix (i.e., P). As such, due to the standardization, symmetry, and ones vector diagonal, the model will solely have to estimate the unique $\frac{k(k-1)}{2}$ inter-correlations in P , which speeds up computation. A more cumbersome alternative would involve fitting $\frac{k(k-1)}{2}$ bivariate linear models to the standardized data matrix, one for each pair of mani-

festations, since the resulting *standardized coefficients* (Wooldridge, 2019: 184-185) are equivalent to the linear inter-correlations (cf. Agresti, 2018: 273).

Irrespective of approach, once the inter-correlations are estimated, they can be used to construct the estimated correlation matrix ($\hat{P}$), which together with its inverse ($\hat{P}^{-1}$), can be used to first compute the anti-image correlation matrix ($\hat{P}_q$) and then the overall ($\hat{\alpha}$) and k individual KMO indices ($\hat{\alpha}_m$). Applying the frequentist formula to the simulated data, a matrix containing the estimated inter-correlations ($\hat{\rho}$), anti-image inter-correlations (i.e. $\hat{\rho}_q$), and the KMO indices, is provided for ease of overview in table 2 in section 6.5 of the Appendix.

The estimated overall KMO index of this simulated data ($\hat{\alpha}$) is .846, which is indicative of a relatively small diffusion in the correlations, consistent with relatively few latent factors which would make it appropriate for a factor analysis. This demonstrates that by calculating the KMO index, the researcher can obtain useful information that enables them to consider the 'sampling adequacy' of the data matrix to be fit for an EFA and to expect few latent factors. By now, the reader should be familiar with the KMO index, its calculation, frequentist conceptualization, interpretation and usefulness, and this review thus turns to a discussion of its inferential viability and robustness.

2.5 Inference

While inferential statements are seldom made in relation to the KMO index, these can be useful to express uncertainty about the estimated 'sampling adequacy' of a data matrix. The absence of such practices may be due to the lack of studies on the sampling distribution of the KMO index, making researchers unable to make inferential statements with reference to a known parametric distribution. However, researchers with sufficiently large and representative samples can generally overcome such a limitation (cf. Agresti, 2018: 140) with the *non-parametric bootstrap* (Efron, 1979, 2003; Efron and Tibshirani, 1994). Here, the data matrix ($\mathcal{D}$) can be permuted a sufficiently large number of times (i.e., the rows of $\mathcal{D}$ are re-sampled with replacement) to produce a distribution of counterfactually sampled data matrices. From each permuted data matrix, an overall and k individual KMO indices (one for each manifestation) can be calculated by applying the functions in section 2.2, to produce an approximation of the sampling distribution of the KMO indices. Probability statements subject to the limitations of Frequentism and the likelihood-based approach (cf. Cohen, 1994; Kruschke, 2014; Wagenmakers et al., 2010) can then be made, for example, in the form of p -values (cf. Stock and Watson, 2019: 110-117) and/or 95% *confidence intervals* (Stock and Watson, 2019: 117-118). While such inferential statements may *not* necessarily be appropriate within the context of an *exploratory* factor analysis (EFA), because such statements theoretically are *confirmatory* and assume *a priori* specified hypotheses, they could be considered useful in relation to a *confirmatory factor analysis* (CFA, Brown, 2015; Levy and Mislevy, 2020), where prespecified hypotheses of the 'sampling adequacy' of a data matrix can be tested prior to testing hypotheses about factor loadings and model structure. Since the 'sampling adequacy' matters just as much for a CFA as for an EFA, such hypotheses are relevant for applied research, and the evidence provided by this procedure can corroborate the frequentist conclusion that the data matrix is factorable at the population level and appropriate for a factor analysis.

An example of applying the frequentist *Null Hypothesis Significance Testing* procedure (NHST, cf. Field, 2018: 72-90; Kruschke, 2014: 300-333) to make inferential statements about the KMO index can be demonstrated for the previous example. Here, the *null hypothesis* (H_0) can be stated as $\text{KMO}_{\text{overall}} = .5$, which reflects the aforementioned KMO index value that can be considered 'indiscernible' in relation to an assessment of 'sampling

adequacy' (see Section 2.3), and researchers should preferably be able to eliminate the uncertainty expressed by this demonstrably plausible value. As such, the *alternative hypothesis* (H_a) will state that $KMO_{\text{overall}} \neq .5$, and the rejection level for H_0 will be prespecified at the conventional 5% level. Here, the mean, *bias-reduced* (Park et al., 2022; Park and Wang, 2020, 2022) *standard error* (SE, Stock and Watson, 2019: 113-114), and 95% *bias-corrected and accelerated interval* (BCa, Efron, 1987) are calculated from 40,005 permutations of the data. This number of permutations was chosen to help invoke the *law of large numbers* (Angrist and Pischke, 2015: 13-16; Imai, 2017: 300-302; Sen and Singer, 1993; Stock and Watson, 2019: 85-86) and for comparability with a later demonstration of the Bayesian KMO index (see Section 3.3). This results in an estimated overall KMO index ($\hat{\alpha}$) of .831 (SE = .012; 95% BCa[.805; .851]), where the SE and 95% BCa, given their frequentist conceptualization, help express the (sampling) uncertainty about the previously conceptualized *estimated 'sampling adequacy' in the population* ($\hat{\alpha}$). Since the estimated sampling distribution of the KMO index is with 95% confidence higher than a value of .5, this is consistent with the sampling adequacy being discernible enough for a proper assessment, and the relatively high values are also consistent with a low level of diffusion in the correlations at the population level, suggesting that the data matrix is factorable and appropriate for a subsequent factor analysis.

2.6 Robustness

With the inferential viability of the KMO index having been demonstrated, this section clarifies why the arguably strict stipulations were necessary when specifying the data matrix in section 2.1. While those stipulations by definition hold for data simulated under such specifications, for any real-world data, they merely constitute assumptions that may be violated. Should this be the case, the conclusions regarding the 'sampling adequacy' of a data matrix provided by the KMO index may be incorrect.

As made evident by the formula in section 2.2, the KMO index is a function of the correlation matrix. Because the calculation of the correlation matrix can be considered a special application of the GLM, the robustness of the KMO index is directly tied to the *Gauss-Markov theorem* (Stock and Watson, 2019: 103-104, 195; Wooldridge, 2019: 88). Besides requiring the manifestations to be measurable on an *interval scale*, be meaningfully reducible to *quantitative* constructs (Agresti, 2018: 24; Field, 2018: 10-13), and have no missing values (as discussed in section 2.2.6), this means that the KMO index generally makes assumptions that include the following:

Random sampling (Agresti, 2018: 30; Stock and Watson, 2019: 158-161, 176; Wooldridge, 2019: 41, 80): If one wishes to credibly infer results about the 'sampling adequacy' from a sample to a population, the probability of the units being included in the sample must effectively be equivalent to their proportion in the population. As used in the data specification in sections 2.1 and 2.4, this is generally best achieved with random sampling. Violating this assumption could bias inference if the sampled units are systematically different from the unit population (i.e., *sampling bias*, Agresti, 2018: 30-31), possibly resulting in incorrect inference (i.e., a *Type III* error, Gigerenzer, 2004: 599). For example, if the correlations used to compute the KMO index suffer from a wrongly estimated magnitude and direction (i.e., a *Type M* or *Type S* error, respectively, cf. Gelman et al., 2021: 59; Gelman and Carlin, 2014). In real-world scenarios, there are no easy solutions to a violation of this assumption, though common approaches involve re-weighting data to better match the population (e.g., Downes et al., 2018), which could improve the inferential viability of the KMO index.

Sample variation (Wooldridge, 2019: 42): None of the measured manifestations can be constant, that is,

$\forall \theta_m : \text{Var}(\theta_m) > 0$). This is because the linear correlation is only defined for *variable* phenomena, as the formulas in function (2.2.5) otherwise include a division by zero. For that reason, sections 2.1 and 2.4 specified a heterogenous population and large sample, with positive standard deviations for both the latent and manifest phenomena. This assumption can easily be assessed by checking whether all of the measured manifestations vary, and a violation could be solved by collecting more data but can otherwise be difficult to solve.

No perfect multicollinearity (Stock and Watson, 2019: 226-231; Wooldridge, 2019: 80): There is no exact linear association between any manifestation, because, as previously noted in 2.2, the correlation matrix will otherwise be singular and *not* invertible, making the KMO index undefined. If the data matrix is full column rank, which was specified in section 2.1, this assumption is met by definition. As demonstrated in section 2.4, multicollinearity can be assessed with the VIF. If it is violated, a solution may be to identify and remove the manifestations responsible for the multicollinearity from the analysis.

Homoskedasticity (Stock and Watson, 2019: 188-192; Wooldridge, 2019: 45, 88): All of the manifestations must have constant variance as assumed by the variance-covariance matrix (Ω) of the GLM in section 2.4. This was ensured by specifying constant standard deviations for both the latent and manifest phenomena in sections 2.1 and 2.4. Violating this assumption can reduce the ability to infer results, though this can be somewhat mitigated if inference is made with the non-parametric bootstrap used in section 2.5. Alternative solutions could possibly involve modeling instances of *heteroskedasticity* (e.g., with *weighted least squares*, Agresti, 2018: 449; Field, 2018: 239; Stock and Watson, 2019: 195-196; Wooldridge, 2019: 273-283).

Zero conditional mean (Imai, 2017: 370-371; Stock and Watson, 2019: 157-158, 160-161; Wooldridge, 2019: 42, 82-83): None of the manifestations can be measured with systematic error. This assumption would be violated if the manifestations were systematically mismeasured, or if their relationship with the latent phenomenon was *spurious* (Agresti, 2018: 306-308; Gumbel, 1933) or *masked/suppressed* (i.e., *confounded*, Agresti, 2018: 309; McElreath, 2019: 144-153; Lenz and Sahn, 2021), potentially leading to *false positive* or *false negative* conclusions regarding the factorability of the data matrix (i.e., a *Type I* and *Type II* error, respectively, cf. Neyman and Pearson, 1933). In sections 2.1 and 2.4, this was guaranteed by specifying the latent phenomenon to be exogenous of all confounders and temporally preceding the manifestations, which for simplicity, were also specified to occur simultaneously in time. Violations of this assumption are hard to solve and are perhaps best addressed *a priori*, with accurate measures, exogenous manipulation of the latent phenomenon, and sound theory.

Independent and identically distributed (Stock and Watson, 2019: 82): All the units and manifestations must be *iid*, as specified with independent measurements and identical probability distributions in sections 2.1 and 2.4. Violating this assumption can bias inference (Stock and Watson, 2019: 108), with uncertainty estimates often becoming too small, though solutions could possibly include modeling the *clustering* (i.e., interdependence) of the units and/or manifestations with *multilevel* models (e.g., Gelman and Hill, 2007).

No outliers (Stock and Watson, 2019: 176): There can be no values of the measured manifestations in the sample with a sufficiently high *leverage* (Agresti, 2018: 442) to make the inter-correlations unrepresentative of the population of interest. This assumption was met by specifying that the sample was to be sufficiently large, representative, and drawn from the normal probability distribution in sections 2.1 and 2.4. While the leverage of outliers can generally be reduced by increasing the sample size, another approach is to change the likelihood distribution of model (2.4.2) from a normal to a *Student t* (Gosset, 1908; Helmert, 1875, 876a,b; Lüroth, 1876;

Pearson, 895b). This is because the Student t probability distribution, besides containing a mean and standard deviation parameter, includes a *degrees of freedom* parameter (df , Agresti, 2018: 126-132; Imai, 2017: 339-342; McElreath, 2019: 233-234; Stock and Watson, 2019: 80; Wooldridge, 2019: 708-709), which controls the width of the tails of the distribution. The highest width is achieved at the lower bound of 1, while the lowest width is achieved as df approaches infinity, where the Student t becomes equivalent to the normal. By allowing for a greater width of the tails, the Student t can be considered more robust, because is better able to handle outliers (Gelman et al., 2021: 264-286; Juárez and Steel, 2010; Kruschke, 2014: 458-468; McElreath, 2019: 233-234; O'Hagan, 1979), reducing their leverage and ability to influence the results. Since the Student t is seldom implemented for GLMs in the frequentist framework, a Student t multivariate model is *not* implemented in relation to the frequentist KMO index, but given its common use in Bayesian statistics (e.g., Juárez and Steel, 2010; O'Hagan, 1979), it will be discussed in relation to the Bayesian KMO index (see Section 3.5).

Linearity (Wooldridge, 2019: 40, 80): All of the manifestations must be (approximately) linearly related to the latent phenomena, as specified in sections 2.1 and 2.4, since the PPCC is a *linear* measure of association (Imai, 2017: 143; Stock and Watson, 2019: 157). This assumption is typically violated (cf. Kruschke, 2014: 424), for example, if the manifestations have been measured on an *ordinal* scale (Agresti, 2018: 24-25; Field, 2018: 12) instead of an interval scale. Such issues can be mitigated by transforming the measured manifestations into *ranks* (cf. Field, 2018: 284) and applying the aforementioned standardization function (2.4.1) covered in section 2.4. This joint rescaling-procedure, henceforth denoted *rank-standardization*, effectively transforms the PPCC into Spearman's (1904) *rank correlation coefficient* (SRCC, cf. Agresti, 2018: 247-248; Field, 2018: 351-352; Spearman, 1910). Such a rescaling-procedure can be shown by letting $\text{Rank}(\theta_m)$ denote the *ranking function* (cf. Field, 2018: 288; R Core Team, 2025).

$$\text{Rank}(\theta_m) \equiv \frac{1 + \#\{l \mid x_l < \theta_{u,m}\} + \#\{l \mid x_l \leq \theta_{u,m}\}}{2} \quad (2.6.1)$$

where $\#\{l \mid x_l < \theta_{u,m}\}$ expresses the counts less than the u th value of manifestation m , and $\#\{l \mid x_l \leq \theta_{u,m}\}$ expresses the counts less than or equal to the u th value of manifestation m , which together with 1 in the numerator and the division by 2 serves to account for ties through averaging. Then, let $Z_{\text{rank}}(\theta_m)$ denote the *rank-standardization function* of manifestation m , and while its conceptual output, similar to function (2.4.1), varies across statistical approaches, it can be specified irrespective of framework as:

$$Z_{\text{rank}}(\theta_m) \equiv \frac{\text{Rank}(\theta_m)_u - \text{Mean}(\text{Rank}(\theta_m))}{\text{SD}(\text{Rank}(\theta_m))} \quad (2.6.2)$$

Calculating the KMO index on data transformed using function (2.6.2) enables it to capture 'sampling adequacy' in relation to *ordinal* relationships, which unlike the PPCC does *not* require linear relationships between the latent phenomenon and its manifestations. Instead, it merely requires the relationships between the theorized manifestations and the latent phenomenon to be *monotonous* (cf. Clapham and Nicholson, 2014). Researchers may be unsure when to apply such a transformation, but in the event of linearity, SRCC approximates PPCC, so researchers could by default do rank-standardization and compare results with and without this rescaling-procedure.¹¹

¹¹ An alternative to the PPCC and SRCC is Kendall's (1938) τ , which despite providing more conservative estimates of correlation (cf. Field, 2018: 353; Howell, 2012) is harder to meaningfully interpret, and comparability with the aforementioned coefficients is mostly lost (cf. Field, 2018: 363; Strahan, 1982; though see Moran, 1948).

For the above example, rank-standardizing every manifestation in the data matrix and then calculating the KMO index yields an estimated overall KMO ($\hat{\alpha}$) of .840. The negligible difference between this result and the previous linear-dependent result (i.e., $\Delta.006$) can be interpreted as being consistent with linearity of the functional form underlying the relationship between the latent phenomenon and the theorized manifestations.

Taken together, the robustness of the KMO index thus relies on numerous assumptions that were accounted for when specifying the data matrix in sections 2.1 and 2.4. However, for most real-world data, any of these could potentially be violated, possibly resulting in misleading results. Accordingly, to ensure a valid and reliable KMO index, researchers should be wary of violating the stipulations made here. Having revisited the frequentist KMO index, a Bayesian reconceptualization can now be made.

3 The Bayesian KMO index

The first part of this paper (Section 2) revisited the classical KMO index (i.e., 'Little Jiffy, Mark IV', Kaiser and Rice, 1974), thoroughly reviewing its formula, interpretation, viability for inference, and robustness. By considering its calculation and conceptualization within different statistical frameworks, this KMO index was found to be frequentist, leaving it incompatible with Bayesian inference. For the purposes of that review, the term 'MLE-Bayesian' was specifically introduced in section 2.2 to jointly refer to a likelihood-based conceptualization of the KMO index that could be considered Bayesian in the context of 'non-informative' priors and maximum-likelihood point estimators. While this served to build towards a possible Bayesian conceptualization, the formula and concepts specific to that terminology evidently failed to enable the estimation of a posterior distribution and the incorporation of prior information, making it somewhat 'improper' for Bayesian inference.

In this section, the limitations of that MLE-Bayesian approach are solved by introducing an arguably more proper *Bayesian Kaiser-Meyer-Olkin index*. This is accomplished by reconceptualizing the KMO index through use of Bayes' theorem and discussing the specification of priors and estimation of a posterior distribution (Section 3.1), resulting in an adjustment of the formula introduced in section 2.2 to derive the Bayesian KMO (BKMO) index from a posterior of correlations (Section 3.2). This is followed by an example demonstrating the computation of this BKMO index (Section 3.3), how to assess whether results are sensitive to priors (Section 3.4), and a discussion of a robust BKMO index (Section 3.5).

3.1 Reconceptualization

To reconceptualize the KMO index in a 'proper' Bayesian manner, it is necessary to consider *Bayes' theorem* (Bayes and Price, 1763; Laplace, 1814), sometimes known as *Bayes' rule* (e.g., Imai, 2017: 266; Kruschke, 2014: 100-104), which is the conceptual foundation of Bayesian inference (Gelman et al., 2014; Levy and Mislevy, 2020; McElreath, 2019; Kruschke, 2014). As shown in equation (3.1.1) (cf. Field, 2018: 124; Imai, 2017: 266; McElreath, 2019: 49), Bayes' theorem consists of $P(a)$, which is the *prior probability*; $P(b)$, the *marginal likelihood*; $P(b|a)$, the *likelihood*; and $P(a|b)$, the *posterior probability*.

$$P(a|b) = \frac{P(a)P(b|a)}{P(b)} \quad (3.1.1)$$

Within Bayesian inference (Gelman et al., 2014; Levy and Mislevy, 2020; McElreath, 2019; Kruschke, 2014),

a and b are often respectively conceptualized as the *parameter(s)* and the *data*. In equation (3.1.1), this means that $P(a)$ is the (prior) probability of the parameter, $P(b)$ is the marginal likelihood of the data, and $P(b|a)$ is the likelihood of the data given the parameter. While this often leads researchers to conceptualize $P(a|b)$ as the 'probability of the parameter given the data' (e.g., Kruschke, 2014), it is worth emphasizing that this posterior is dependent on the prior, $P(a)$, and likelihood, $P(b|a)$, making conclusions sensitive to their specification. Accordingly, the specification of the prior and likelihood can be taken to reflect *modeling* choices that may impact the posterior, and because researchers could disagree about these choices, they may present a point of contention. To appropriately reflect this dependency on possibly subjective modeling choices, the posterior, $P(a|b)$, will here be referred to as the *modeled probability of the parameter given the data*.

Researchers working within the frequentist and likelihood-based frameworks should be familiar with the likelihood, $P(b|a)$, since this is the only parameter of Bayes' theorem that is used within these frameworks, either implicitly or explicitly, specifically because these approaches require the specification of a likelihood distribution for the statistical models that aim to describe the data (e.g., the normal likelihood for the GLM, cf. Agresti, 2018; Field, 2018; King, 1998; Stock and Watson, 2019; Wooldridge, 2019). The Bayesian framework differs by instead considering the posterior, $P(a|b)$, to be of central interest. To enable its calculation, in addition to specifying a distribution for the likelihood, Bayesian researchers must also specify a distribution for the prior. A historical limitation of this approach is that the marginal likelihood, $P(b)$, also needs to be calculated, which can be difficult, if not practically impossible, except for only the simplest of models (cf. Kruschke, 2014: 115). However, recent advances in *Markov-Chain Monte Carlo* methods (MCMC, Brooks et al., 2011) have solved this limitation by circumventing the need to calculate $P(b)$ through approximation-based solutions (cf. Gelman et al., 2014: 275-308; Kruschke, 2014: 143-191; Levy and Mislevy, 2020: 93-112; McElreath, 2019: 263-296), enabling entirely new applications of Bayesian inference. Given the aforementioned limited information about some of the properties of the KMO index (see Section 2.5), computational difficulties that could arise from a Bayesian reconceptualization can thus be mitigated by relying on the easily applicable MCMC methods.

With Bayes' theorem in mind, a and b will thus here be respectively reconceptualized as the 'sampling adequacy' (i.e., α_{Bayes}) and data matrix ($\mathcal{D}$). Here α_{Bayes} substitutes the MLE-Bayesian concept of the 'sampling adequacy in the data' (a) introduced in section 2.2, since this 'proper' Bayesian conceptualization reflects the 'sampling adequacy *given* the data', where a may *not* equal α_{Bayes} due to the infusion of prior information. While α could generally be used instead of α_{Bayes} , the latter denotation is preferred here to distinguish it from the frequentist concept of α used in section 2. This reconceptualization of a and b in Bayes' theorem is appropriate, since the formula in section 2.2 showed that the 'sampling adequacy' is a function of the inter-correlations (r) and anti-image inter-correlations (r_q), which both are based on the data matrix ($\mathcal{D}$). For similar reasons, the inter-correlations, inverse inter-correlations, and anti-image inter-correlations, when infused with prior information, can be distinctly denoted ρ_{Bayes} , ρ_{Bayes}^{-1} , and $\rho_{q\text{Bayes}}$, respectively.

In equation (3.1.1), $P(a)$ can then be denoted $P(\alpha_{\text{Bayes}})$ and conceptualized as the *prior probability of the 'sampling adequacy'*, which can be more meaningfully interpreted as the uncertainty about the 'sampling adequacy' before observing the data. Such prior uncertainty can be taken to reflect the stochasticity involved in data collection (e.g., random sampling can produce data of varying quality) or fundamental epistemological uncertainty about the phenomenon being studied. $P(b)$ can then be denoted $P(\mathcal{D})$ and becomes the *marginal likelihood of the data*.

matrix, which, as previously mentioned, can be ignored due to the reliance on MCMC methods. With this in mind, $P(b|a)$ becomes $P(\mathcal{D}|\alpha_{\text{Bayes}})$, which reflects the *likelihood of the data matrix given the 'sampling adequacy'*. From this, it follows that the $P(\alpha_{\text{Bayes}}|\mathcal{D})$ can be conceptualized as the *posterior probability of the 'sampling adequacy'*, that is, the modeled probability of the 'sampling adequacy' given the data matrix.

With the reliance on MCMC methods, researchers thus only need to specify the prior of the sampling adequacy, $P(\alpha_{\text{Bayes}})$, and the likelihood of the data given the 'sampling adequacy', $P(\mathcal{D}|\alpha_{\text{Bayes}})$, to arrive at the posterior probability of the 'sampling adequacy', $P(\alpha_{\text{Bayes}}|\mathcal{D})$. However, considering the relatively abstract concept and interpretation of 'sampling adequacy' and the aforementioned confusion regarding its mathematical properties (see Sections 2.3 and 2.5), researchers may *not* find it easy to specify a prior and likelihood distribution for it. These difficulties can be circumvented by recognizing that researchers need *not* necessarily specify a prior and likelihood for the KMO index *per se*, which is contrary to the specification of most statistical models in Bayesian inference (e.g., McElreath, 2019; for an exception, see, e.g., Kruschke, 2014: 226-230). This circumvention is possible because the KMO index can be recognized as not just a function of the correlation matrix, but a *deterministic* function of the correlation matrix, meaning that there is no *additional* stochastic nor epistemological uncertainty in transitioning from a prior on R – through R^{-1} and R_q – to a prior on a and a_m . Stated differently, the prior and likelihood of the correlation matrix *implies* a prior and likelihood of the KMO index. A prior and likelihood for the former alone are therefore sufficient. Estimating the Bayesian KMO index thus becomes a question of applying Bayesian inference in relation to a correlation matrix.

In line with the above application of the GLM for calculating correlations (see Section 2.4), a likelihood useful for deriving the posterior of a correlation matrix, and thus the 'sampling adequacy', is the aforementioned multivariate normal probability distribution ($\mathcal{MVN}$, Gelman et al., 2014: 578, 582). With this distribution in mind for the likelihood, this leaves the question of specifying the priors for the correlation matrix. A natural choice for such a prior is the *Lewandowski-Kurowicka-Joe* probability distribution ($\mathcal{LKJ}$, Gelman et al., 2014: 578, 584; Lewandowski et al., 2009), which is an often used prior for correlation matrices in Bayesian inference (e.g., Gelman et al., 2014: 55-56; Gelman et al., 2021: 123-127; McElreath, 2019: 441-442). This is, in part, because it accounts for correlation matrices having symmetric off-diagonal elements bounded between -1 and 1 and a diagonal that is a ones vector. The priors can be specified with this $\mathcal{LKJ}$ distribution through use of its single positive real scalar *shape* parameter (i.e., η). When $\eta = 1$, the distribution of correlations is flat; when $\eta > 1$, the distribution of correlations is less extreme, approaching an identity matrix at greater values; with $\eta < 1$, the distribution becomes more extreme, approaching a singular correlation matrix at lower values (cf. McElreath, 2019: 442). This means that $\mathcal{LKJ}$ specifies a joint prior for all the correlations, making it easy for researchers to specify priors with many theorized manifestations.

While this $\mathcal{LKJ}$ probability distribution can be considered generally useful in relation to a Bayesian KMO index, it does *not* permit specifying individual priors for the correlation matrix. Since the KMO index is more commonly applied to *exploratory* rather than *confirmatory* factor analysis, researchers are here expected to *not* possess prior information that would enable discriminating between the inter-correlations of manifestations, making the $\mathcal{LKJ}$ prior appropriate for most practical uses. Should researchers nevertheless possess varying prior information about the inter-correlations, instead of a multivariate normal, they can specify $\frac{k(k-1)}{2}$ individual bivariate linear models with different priors to account for that information when estimating the inter-correlations, as mentioned

in section 2.4.

To reflect the expected lack of substantial prior information about the inter-correlations, researchers should ideally specify the shape parameter to produce a 'weak' or 'non-informative' prior (cf. Gelman et al., 2014: 51-56). Here, in line with Gelman and colleagues (2014: 51-52), the author recommends specifying the former. While a 'non-informative' uniform prior can be specified for the $\mathcal{LKJ}$ with $\eta = 1$, uniform distributions are often *not* considered 'proper' (cf. Gelman et al., 2014: 52), and in real-world applications, there is almost always some prior information that can be extrapolated from existing research and applied to the particular research context. For example, in relation to prior information on correlations between novel phenomena, research on inter-correlations in the literature can be used to create *realistic* prior expectations about these inter-correlations (e.g., Funder and Ozer, 2019; Lovakov and Agadullina, 2021; Gignac and Szodorai, 2016). Should researchers nevertheless prefer 'non-informative' priors, caution is advised, since a uniform distribution is *not* always the correct choice (cf. Gelman et al., 2014: 51-55; Jeffreys, 1946; Kruschke, 2014: 293; Lee and Webb, 2005; Zhu and Lu, 2004).

Specifying 'weakly-informative' priors with the $\mathcal{LKJ}$ probability distribution can be done in numerous ways. One general approach involves specifying 'skeptical, yet persuadable' priors, whose distributions are deliberately conservative by expecting small inter-correlations but which remain wide enough to 'let data speak for itself' (e.g., $\eta = 2$, McElreath, 2019: 441-442). Such priors can impact data in a manner similar to *regularization* by reducing *overfitting*, which can improve the generalization of results (cf. Kruschke, 2014: 531-532; McElreath, 2019: 214-217). Another approach (cf. Kruschke, 2014: 294) could take advantage of the fact that 'tomorrow's prior is today's posterior' (Lindley, 1972: 2) by specifying a prior for each individual inter-correlation using existing results. Such priors could be derived from previous posteriors, or in a manner similar to *cross-validation* (cf. Stone, 1974), by deriving the posterior from a relatively small subset of the current data (e.g., 10% - 20%), and then using this posterior to specify priors for an analysis of the *remaining* data (for a discussion, see, e.g., Berger and Pericchi, 2001). However, in the event that researchers do possess quality prior information, they should *not* stray from using this information to specify 'strongly informative' priors (cf. Kruschke, 2014: 113-115), though they should always transparently report their priors to open them up to criticism from their peers.

Regardless of approach, researchers may be concerned whether their choice of priors for the $\mathcal{LKJ}$ inadvertently influence the results. As a response to this, it can be stated that for most statistical models, as long as the prior assigns non-zero probability to every possible parameter value, data can quickly overwhelm the prior (cf. Kruschke, 2014: 293; Wagenmakers et al., 2010: 167), producing relatively prior-insensitive results that closely resemble frequentist and likelihood-based estimates but for whom a Bayesian interpretation can be made. Similarly, should researchers specify priors using a uniform distribution (e.g., $\eta = 1$, McElreath, 2019: 441-442), results approximate likelihood estimates (cf. Hastie et al., 2017: 272) even with little data (cf. Kruschke, 2014: 113). Researchers can always compare results based on 'strong' or 'weakly informative' priors to results based on 'non-informative' priors, or results derived from the non-parametric bootstrap, to assess whether results are robust to the prior specification, though researchers should note that the re-sampling involved in the non-parametric bootstrap is *not* equivalent to the MCMC method of Bayesian inference (cf. Kruschke, 2014: 161).

With these considerations for the specification of the prior in mind, the *Bayesian* KMO (BKMO) index is thus conceptualized as 'a measure of sampling adequacy *given* the data'. Borrowing denotation from the frequentist approach to distinguish this concept from the MLE-Bayesian approach, this *modeled 'sampling adequacy' given*

the data matrix can be denoted α_{Bayes} , or alternatively α , depending on the need to distinguish it from its frequentist and likelihood-based conceptualizations. Since the Bayesian approach taken here involves *estimating* the posterior 'sampling adequacy' with MCMC methods, the resulting quantity is the *estimated modeled 'sampling adequacy' given the data matrix* (i.e., $\hat{\alpha}_{\text{Bayes}}$, or $\hat{\alpha}$). This is a relevant distinction, since the two quantities (i.e., α_{Bayes} and $\hat{\alpha}_{\text{Bayes}}$) may *not* necessarily be equivalent. For the same reasons, notation is introduced to denote the *estimated* Bayesian inter-correlations (i.e., $\hat{\rho}_{\text{Bayes}}$), inverse inter-correlations (i.e., $\hat{\rho}_{\text{Bayes}}^{-1}$), and anti-image inter-correlations (i.e., $\hat{\rho}_{q\text{Bayes}}$).

3.2 Formula

By now, it should have been made clear that the BKMO index is only indirectly computed using MCMC methods, and instead are to be calculated directly from the posterior distribution of correlation matrices, which is why the likelihood and prior solely needed to be specified for the correlation matrix. This means that the BKMO index can be calculated, analogous to the bootstrapped frequentist KMO index, simply by, for each posterior draw, constructing a correlation matrix from the MCMC-estimated inter-correlations ($\hat{\rho}_{\text{Bayes}}$), which together with the estimated anti-image inter-correlations ($\hat{\rho}_{q\text{Bayes}}$), can be used to estimate the posterior of the overall ($\hat{\alpha}_{\text{Bayes}}$) and individual KMO indices ($\hat{\alpha}_{\text{Bayes}[m]}$) using the MLE-Bayesian formula in functions (2.2.7) and (2.2.8) from section 2.2.

The reliance on these MLE-Bayesian point-estimating formula is appropriate, because all prior information has already been incorporated into the estimated posterior correlation matrix (i.e., $\hat{P}_{\text{Bayes}}$). Accordingly, the formula adjusted for the Bayesian reconceptualization can be concisely expressed by adapting the MLE-Bayesian formula (2.2.7.3) in function (2.2.7), resulting in the 'proper' Bayesian formula (3.2.1), where d denotes the index for the d th posterior draw.

$$\text{BKMO}(\mathcal{D}) \equiv \hat{\alpha}_{\text{Bayes}[d]} \equiv \frac{\sum_{i \neq j} \hat{\rho}_{\text{Bayes}[i,j,d]}^2}{\sum_{i \neq j} \hat{\rho}_{\text{Bayes}[i,j,d]}^2 + \sum_{i \neq j} \hat{\rho}_{q\text{Bayes}[i,j,d]}^2} \quad (3.2.1)$$

Similarly, adapting the MLE-Bayesian formula (2.2.8.3) in function (2.2.8) using the same Bayesian reconceptualization yields the 'proper' Bayesian formula for estimating the posterior 'sampling adequacy' of manifestation m , shown in function (3.2.2).

$$\text{BKMO}(\mathcal{D}_m) \equiv \hat{\alpha}_{\text{Bayes}[m,d]} \equiv \frac{\sum_{j, j \neq i} \hat{\rho}_{\text{Bayes}[i,j,d]}^2}{\sum_{j, j \neq i} \hat{\rho}_{\text{Bayes}[i,j,d]}^2 + \sum_{j, j \neq i} \hat{\rho}_{q\text{Bayes}[i,j,d]}^2} \quad (3.2.2)$$

In accordance with this Bayesian conceptualization of the KMO index, from the distribution of its estimated posterior values, coherent probabilistic statements can be made about the modeled 'sampling adequacy' given the data. For example, one can ascertain the posterior probability that the 'sampling adequacy' is greater than .5 (i.e., $P(\hat{\alpha}_{\text{Bayes}} > .5 \mid \mathcal{D})$), or one can compute the 95% *highest density (contiguous) interval* (HD(C)I, Kay, 024a,b; Kruschke, 2014: 28; Makowski et al., 019b; McElreath, 2019: 56-58), which contains the 95% most probable values of the 'sampling adequacy' given the data, both of which are coherent probabilistic statements within the Bayesian framework (cf. Clayton, 2021; Kruschke, 2014).

The inferential viability of this Bayesian conceptualization is different from the frequentist KMO index, which

cannot incorporate prior information, is highly sensitive to sampling intentions, and where probabilistic statements are less intuitive, since the 'sampling adequacy' is fixed for any particular sample, varies solely across samples, and are about the data given the 'sampling adequacy' (i.e., $P(\mathcal{D}|\alpha)$). By enabling the typically more interesting statements about the 'sampling adequacy' given the data (i.e., $P(\alpha|\mathcal{D})$, cf. Clayton, 2021; Kruschke, 2014; McElreath, 2019), the Bayesian conceptualization arguably provides better inferential viability than its frequentist counterpart. Having thus reconceptualized the KMO index within the Bayesian statistical framework and introduced it as the *Bayesian KMO* (BKMO) *index*, a demonstration is provided.

3.3 Example

To demonstrate the conceptualized Bayesian KMO index, the simulated data matrix previously specified (see Section 2.4) is once again used. Contrary to the frequentist approach, the manifestations are standardized using the MLE-Bayesian formula for function (2.4.1), which is preferred to a 'proper' Bayesian standardization for computational ease, since the the estimation would otherwise have to account for a posterior of standardizations as input. Using the same notation introduced in section 2.4 (cf. Levy and Mislevy, 2020; McElreath, 2019: 441-442), a *Bayesian generalized linear model* (BGLM) is formalized.

$$Z(\theta)_u | \mu_m, \Omega \sim \mathcal{MVN}(\mu_m, \Omega) \text{ for } u \text{ in } 1, \dots, n \quad (3.3.1)$$

$$\Omega = DPD$$

$$\mu_m = 0 \text{ for } m \text{ in } 1, \dots, k$$

$$\text{diag}(D)_m = 1 \text{ for } m \text{ in } 1, \dots, k$$

$$P \sim \mathcal{LKJ}(2)$$

This model (3.3.1) states that the individual standardized manifestations (i.e., $Z(\theta)_u$) in the data matrix ($\mathcal{D}$) are *iid*¹² as a conditionally multivariate normal distribution ($\mathcal{MVN}$), with a k -length manifestation-varying mean vector parameter (i.e., μ_m) and a k times k variance-covariance matrix (i.e., Ω) parameter. Similar to the GLM in section 2.4 (cf. Barnard et al., 2000), the variance-covariance matrix can be decomposed into two k times k diagonal matrices of standard deviations (i.e., D) and one k times k correlation matrix (i.e., P). Given the standardization of each manifestation in $\mathcal{D}$ and lack of predictors, the mean vector (μ_m) becomes a zero vector of intercepts, while D simplifies to an identity matrix. This reduces the variance-covariance matrix to the correlation matrix (P), whose unique $\frac{k(k-1)}{2}$ inter-correlations become the only estimands of the model for which priors will have to be specified. The prior distribution of P is specified using the $\mathcal{LKJ}$ distribution (Lewandowski et al., 2009), whose shape parameter (i.e., η), consistent with recommendations by McElreath (2019: 442), is specified as 2 to reflect an *a priori* belief in relatively small inter-correlations, which is intended to be 'regularizing/weakly informative' relative to the data (cf. McElreath, 2019: 214-217, 442; Gelman et al., 2014: 55-56; Gelman et al., 2021: 123-127).

¹²Consistent with the Bayesian framework (Gelman et al., 2014; Levy and Mislevy, 2020; McElreath, 2019), the relatively stringent *iid* assumption in model (3.3.1) may be relaxed by replacing it with the weaker assumption of *exchangeability* (de Finetti, 1974). This means that the units could be interdependent (e.g., by belonging to distinct clusters), but that the priors and model treat them as exchangeable given the available information (cf. Levy and Mislevy, 2020: 56-64; McElreath, 2019: 81).

To visualize the effective prior for the 'sampling adequacy' that results from this prior for the correlation matrix, MCMC can be used to sample from the prior distribution of model (3.3.1) (cf. Kruschke, 2014: 212). The effective prior distribution of the 'sampling adequacy' resulting from this $\mathcal{LKJ}$ prior has an average of .304 (SD = .093; 95% HDI [.127; .480]), with the estimated distribution being shown in figure 1. Inspecting this prior for the 'sampling adequacy' reveals an *a priori* belief in a relatively low 'sampling adequacy'. This can be substantively interpreted to reflect a conservative expectation that the data will be relatively *infactorable* due to a high diffusion in correlations and thus *inappropriate* for factor analysis. In this instance, a shape parameter for the $\mathcal{LKJ}$ prior distribution of 2 for 10 theorized manifestations effectively implies a 'skeptical' prior for the 'sampling adequacy'. While the width of this distribution can be taken to indicate that it is also *a priori* 'persuadable', this property of the prior is ultimately determined in relation to the strength of the data, and as such, whether the prior is 'persuadable' will be assessed through comparisons with results derived from a 'non-informative' prior.

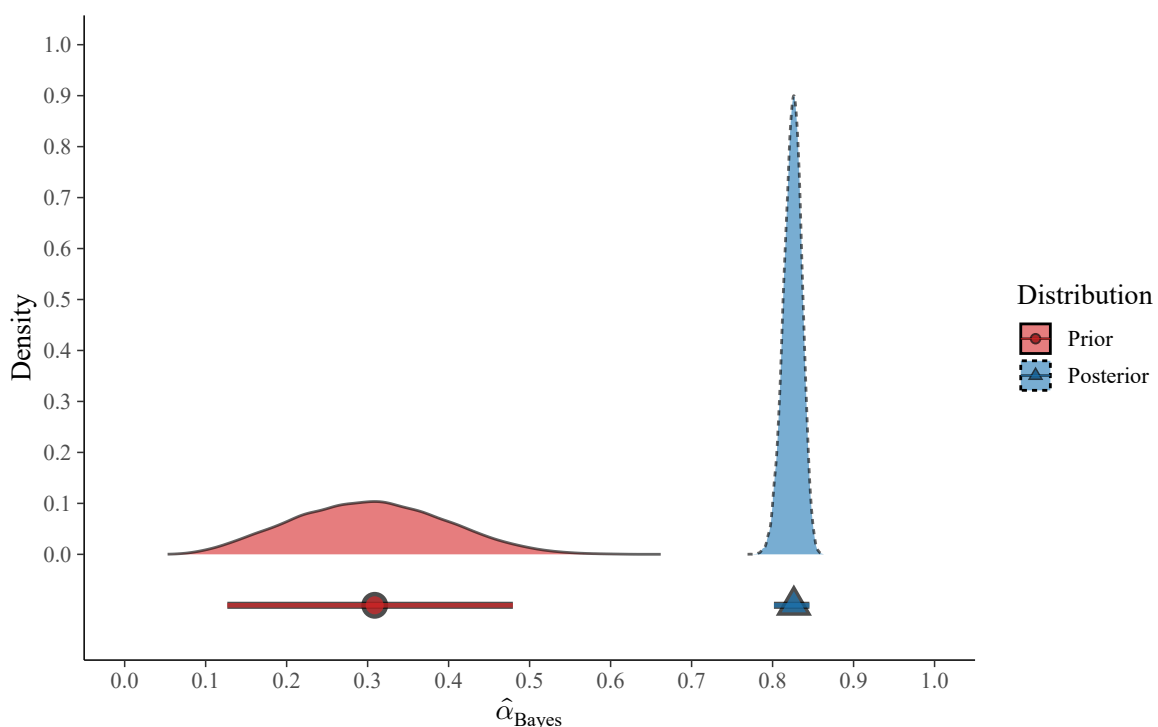


Figure 1. Overall 'sampling adequacy'

NOTE: Prior/posterior distributions of the estimated overall Bayesian KMO index (i.e., $\hat{\alpha}_{\text{Bayes}}$). Prior/posterior samples = 40,005. The geometrics below the distributions indicate the mode (circle) and the 95% HDI (bar). Results derived from simulated data.

Model (3.3.1) is then fit to the simulated data using the *brms* R package (Bürkner, 2017, 2018), which is useful due to its flexibility in model specification and reliance on the *STAN probabilistic programming language* (Stan Development Team, 2025a) that enables efficient sampling from the posterior using the *No-U-Turn Sampler* (NUTS, Hoffman and Gelman, 2014) for Hamiltonian MCMC (cf. Kruschke, 2014: 400-416). To help assess whether the sampler converged to the posterior distribution, 15 independent sampling chains were used, and to improve estimation reliability, 2,000 warm-up samples were drawn to increase the chances of convergence prior to sampling. After the warm-up period, 4,667 samples were drawn from each chain, resulting in a total of 40,005 posterior samples, whose size was chosen to help invoke the law of large numbers (Angrist and Pischke, 2015;

Imai, 2017: 300-302; Sen and Singer, 1993; Stock and Watson, 2019: 85-86), as well as ensure a sufficient *effective sample size* (i.e., $ESS \geq 10,000$, Bürkner, 2017; Kass et al., 1998; Kruschke, 2014: 182-184) after accounting for *autocorrelation* (Kruschke, 2014: 182). Based on visual inspections of *trace plots* (Kruschke, 2014: 179-180), adequate values of the *Gelman-Rubin convergence metric* ($\hat{R} < 1.01$, cf. Vehtari et al., 2021; see also Gelman and Rubin, 1992) and the *Monte Carlo Standard Error* (i.e., $MCSE \leq .001$, cf. Kruschke, 2014: 186-187), the sampling chains showed signs of convergence, suggesting the posterior distribution had been reliably identified.

Having estimated the posterior distribution of the inter-correlations matrix, the BKMO index can then be calculated, using the formula introduced in section 3.2, by constructing a correlation matrix from the estimated correlation coefficients for each posterior draw. Since this estimated correlation matrix is infused with prior information, it is no longer denoted R as previously conceptualized in section 2.2 for the MLE-Bayesian approach. Instead, like the other Bayesian metrics, it is the estimated modeled correlation matrix given the data, which can be denoted $\hat{P}_{\text{Bayes}}$ to distinguish it from the frequentist conceptualization, though it may also be denoted $\hat{P}$ when the context is clear. This is followed by calculating the estimated modeled anti-image correlation matrix given the data (i.e., $\hat{P}_{q[\text{Bayes}]}$, alternatively $\hat{P}_q$) from the inverse estimated modeled correlation matrix given the data (i.e., $\hat{P}_{\text{Bayes}}^{-1}$, alternatively $\hat{P}^{-1}$), using the MLE-Bayesian formula (2.2.6.3) in function (2.2.6) for each posterior draw. From the resulting posterior distribution of matrices, the overall and individual BKMO indices are calculated using the Bayesian formula (3.2.1) and (3.2.2), respectively. Since the posterior distribution consists of 40,005 samples, the function implementing the calculation of the BKMO relied on *vectorization* to speed up computation (cf. Kruschke, 2014: 408). This BKMO index function was implemented using R code and is provided for ease of implementation in section 6.2 of the Appendix.

The estimated posterior distribution of the BKMO index, that is, the estimated modeled factorability given the data matrix ($\hat{\alpha}_{\text{Bayes}}$) is provided visually next to the prior distribution in figure 1. These results can be summarized metrically, with the posterior average of the overall BKMO index being .825, with a posterior standard deviation¹³ (SD) of .011. The 95% HDI reveals that the ‘sampling adequacy’ with 95% credibility is between .802 and .845 given the data. Comparing the posterior to the prior distribution reveals an overlap of .000% and an average difference of .521 (SD = .093; 95% HDI[.346; .701]), revealing that the two distributions are credibly different. The fact that the credible values of the ‘sampling adequacy’ excludes .5 and have become substantially higher than this value is consistent with a discernible and credibly high ‘sampling adequacy’. This means that the diffusion in the correlations are relatively low with a high degree of credibility, and the data matrix could thus be considered factorable and appropriate for a subsequent factor analysis.

3.4 Prior sensitivity

To demonstrate how researchers can assess the sensitivity of results to their choice of priors, the posterior derived from strongly or weakly informative priors can be compared to results derived from ‘non-informative’ priors or bootstrapped-based approaches.

The first approach can be demonstrated here by respecifying model (3.3.1) so that the $\mathcal{LKJ}$ prior has a shape parameter of 1, creating a flat prior probability for the correlation matrix. Refitting it using the same MCMC software yields similarly reliable model diagnostics, which are *not* reported here for brevity. Since the priors

¹³Consistent with the MLE-Bayesian approach, the standard deviation of the posterior is calculated using formula (2.2.3.3) for function (2.2.3) (see Section 2.2).

of this model are 'non-informative', this model approximates a maximum-likelihood model, and the resulting 'likelihood' distribution is illustrated in figure 2, where the average BKMO index value is .825 (SD = .011; 95% HDI[.802; .845]). The overlap between this 'likelihood' distribution and the posterior is approximately 100%, with an average difference of .000 (SD = .000; 95% HDI[.000; .000]), revealing them to be credibly *indifferent*. This corroborates the claim that the prior specified in model (3.3.1) is 'persuadable'/'weakly informative' and showcases that the BKMO can approximate likelihood-based results with enough data.

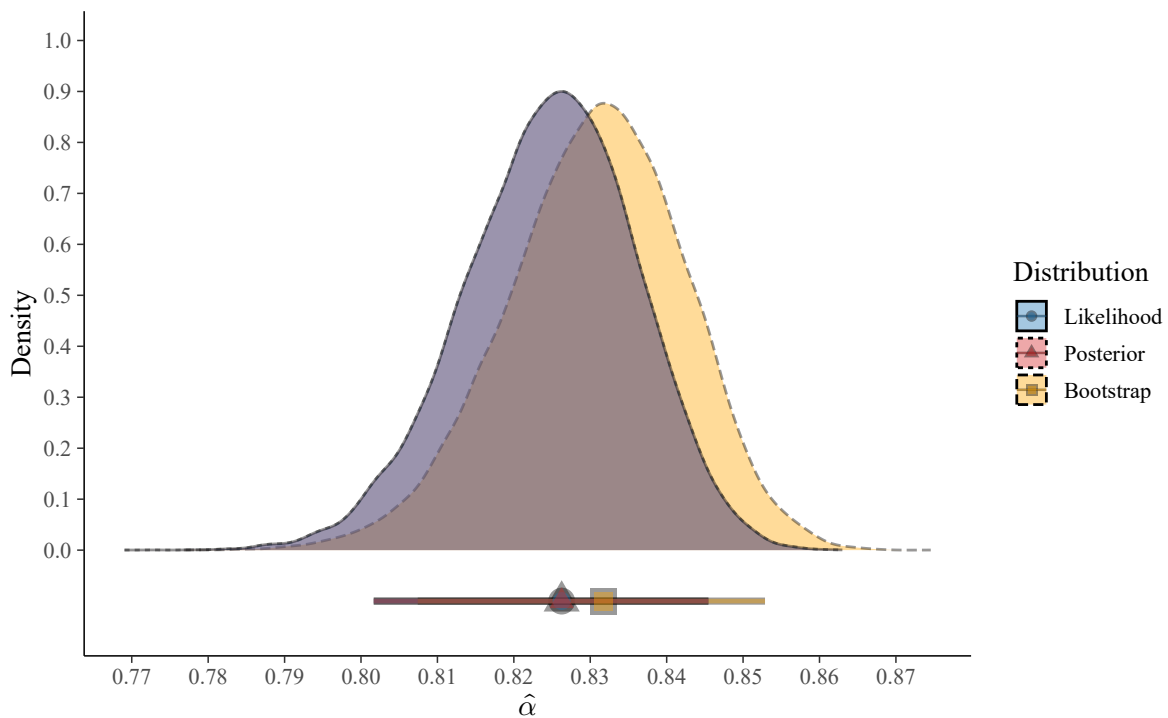


Figure 2. Overall 'sampling adequacy'

NOTE: Likelihood, posterior, and (non-parametric) bootstrap distributions of the estimated overall KMO index (i.e., $\hat{\alpha}$). Likelihood/posterior/bootstrap samples/permutations = 40,005. The geometrics below the distributions indicate the mode (circle) and the 95% HDI (bar). Results derived from simulated data.

While a Bayesian model with 'non-informative' priors can often be fitted using MCMC methods, sometimes for complex models, it can be difficult to get the model to reliably converge. An alternative approach involves a comparison with results from the non-parametric bootstrap, which similarly approximates MCMC with 'non-informative' priors. This model was fit previously when discussing the inferential viability of the frequentist KMO index in section 2.5, and the resulting (non-parametric) bootstrap distribution is also provided in figure 2. The results of this model are *not* repeated here, but the overlap between this bootstrap distribution and the posterior is approximately 79.4%, with an average difference of -.006 (SD = .016; 95% HDI[-.038; .026]). This relatively high degree of overlap and lack of a credible difference is again consistent with a 'weakly informative' prior, and it showcases that the BKMO can approximate a bootstrapped frequentist KMO index.

As such, the similarities in results across the Bayesian, likelihood-based, and bootstrapped frequentist approaches reflect the aforementioned properties of Bayesian inference, where 'weakly informative' priors can be quickly overwhelmed by data, resulting in the convergence of results. Results from Bayesian inference can thus

approximate frequentist/likelihood-based results, though solely the Bayesian-based approach enables the incorporation of prior information and coherent probabilistic statements from the posterior.

3.5 Robustness

Like its frequentist counterpart, all of the assumptions discussed in section 2.6 apply to the BKMO index. Fortunately, it can be made similarly robust to violations of the assumptions of linearity and the presence of outliers. Because the data matrix was specified and simulated in sections 2.1 and 2.4 to ensure linearity and no outliers, and the fact that frequentist and Bayesian approaches have already been demonstrated to provide similar results, for brevity, the implementation of a robust BKMO index will *not* be demonstrated here. Instead it will only be conceptualized and discussed.

As previously discussed (see Section 2.6), if the manifestations exhibit non-linearity or are measured with an ordinal scale, the manifestations can be rank-standardized to reduce the influence of outliers and identify ordinal relationships, making the resulting KMO index more robust to these violations. Consistent with the Bayesian framework, a robust BKMO index could be achieved by simply rank-standardizing the data matrix using the MLE-Bayesian formula for function (2.6.2) before fitting model (3.3.1), followed by computing the BKMO index in line with section 3.3.

While rank-standardizing is one way to derive a robust BKMO index, a more common Bayesian approach, previously alluded to in section 2.6, involves changing the likelihood of the multivariate model (3.3.1) from the normal to the more outlier-robust Student t (Gelman et al., 2021: 264-286, 437; Juárez and Steel, 2010; Kruschke, 2014: 458-468; McElreath, 2019: 233-234; O'Hagan, 1979). By basing the likelihood of the model on a *multivariate Student t probability distribution* ($\mathcal{MV}\mathcal{T}$, Gelman et al., 2014: 580), this introduces a *normality* (i.e., ν) parameter¹⁴ (Kruschke, 2014: 458-459) that also requires a prior specification. At the lower bound of 1, the Student t is equivalent to a *Cauchy* probability distribution (Li and Nadarajah, 2020), for which the mean and variance are *not* defined (being defined at a ν of 2 and 3, respectively), and as previously mentioned, as ν approaches infinity, the Student t becomes equivalent to the normal probability distribution (Gelman et al., 2021: 264-286, 437; Stock and Watson, 2019: 80; Wooldridge, 2019: 708-709).

A common Bayesian approach to specifying the prior for ν (e.g., Juárez and Steel, 2010; Simpson et al., 2017) involves the *natural logarithmic* function (i.e., $\ln(\cdot)$, Imai, 2017: 91-92) and the *gamma probability distribution* ($\mathcal{G}$, Agresti, 2018: 449-450; Gelman et al., 2014: 578, 583; Kruschke, 2014: 236-245; McElreath, 2019: 315), whose joint application ensure that the prior for ν accounts for its lower bound of 1. This gamma distribution can be characterized by a *shape* and *rate* parameter (Kruschke, 2014: 236-245), and Juárez and Steel (2010) find that a shape parameter of 2 and a rate of 0.1 have general advantages when estimating the model. Here, a robust BKMO index could thus be achieved by respecifying the likelihood of model 3.3.1 to: $Z(\theta)_u | \mu_m, \Omega \sim \mathcal{MV}\mathcal{T}_{\nu_u}(\mu_m, \Omega)$ for u in 1, ..., n ; and appending the Gamma specification for the prior at the end of the model: $\ln(\nu)_u \sim \mathcal{G}(2, 0.1)$. Fitting that model would enable the inter-correlations to be estimated with greater robustness, yielding a more robust BKMO index in turn.

¹⁴In the frequentist framework (Agresti, 2018: 270; Field, 2018: 59; Imai, 2017: 168-169, 339-342; Stock and Watson, 2019: 154; Wooldridge, 2019: 94), ν is referred to as a *degrees of freedom* parameter (*df*), which is the number of parameters that are free to vary, being the difference between the sample size (i.e., n) and the number of estimated parameters (i.e., p). In that approach, ν must be at least 0 for the model to be identifiable and at least 1 to be able to assess uncertainty about the model. This is different from the Bayesian approach, where models technically can be identified at a *df* less than zero due to the reliance on priors.

4 Conclusion

This paper has introduced the Bayesian KMO index. This was accomplished by reviewing the classical Kaiser-Meyer-Olkin (KMO) index, which is a Measure of Sampling Adequacy (MSA) that helps assess whether a data matrix is appropriate for a factor analysis. To build towards a Bayesian KMO index (Section 2), this review revisited its calculation and discussed its mathematical and conceptual properties, which served to show that the KMO index is a frequentist statistic, whose inferential viability and robustness rely on strict assumptions about the data matrix. Bayes' theorem was then used to reconceptualize the KMO index (Section 3), resulting in a Bayesian KMO (BKMO) index that can incorporate prior information and enable coherent probabilistic statements to be made about the 'sampling adequacy' given the data. Estimated with MCMC methods, its ease of computation was demonstrated, and this paper provides efficient code to easily implement it in R software.

The conceptualization of a BKMO index constitutes a novel contribution to the literature on statistical inference, in particular, the subdiscipline of factor analysis. This is because it enables researchers to incorporate prior information when assessing the 'sampling adequacy' of a data matrix, and from the resulting posterior distributions, coherent probabilistic statements about the factorability of the data matrix can be made. To remain consistent within their statistical framework, Bayesian researchers should ideally substitute the frequentist KMO index for its Bayesian counterpart whenever they are interested in assessing the 'sampling adequacy' prior to a factor analysis. The areas of application for the BKMO index are numerous, especially within the *psychometrics* subdiscipline of psychology (e.g., Brown, 2015; Levy and Mislevy, 2020), where it can provide critical information to researchers developing new measurement scales for latent phenomena. Researchers can assess the sensitivity of results to their priors through comparisons to results based on uniform priors or bootstrapped-based approaches. Like its frequentist counterpart, the BKMO can also be made more robust to violations of linearity and the presence of outliers, for example, by rank-standardizing the data matrix prior to the analysis or by using a Student *t* multivariate likelihood for the statistical model. The usefulness of the BKMO index is *not* restricted to exploratory factor analyses, and can be extended to confirmatory factor analyses, where researchers can prespecify and test hypotheses about the sampling adequacy. As such, the BKMO index is a novel and widely applicable tool for Bayesian researchers that can offer relevant insights prior to a factor analysis.

5 Acknowledgements

This study was made possible by building on insights and contributions from numerous researchers across disciplines, in particular Henry F. Kaiser, Edward P. Meyer, Ingram Olkin, and John Rice. The author would like to thank those who provided helpful comments on the paper and the staff of Loyola University, Chicago, for providing clarifying information about the authors of the Kaiser-Meyer-Olkin index.

References

- Agresti, A. (2018). *Statistical Methods for the Social Sciences* (5 ed.). Pearson. Global Edition.
- Ahmad, U. S., S. Safdar, and M. Azam (2024). An assessment of bilateral debt swap financing indispensable for economic growth and environment sustainability: a policy implication for heavily indebted countries. *Environmental Science and Pollution Research International* 31(4), 5716–5734.

- Albers, C. and D. Lakens (2018). When power analyses based on pilot data are biased: Inaccurate effect size estimators and follow-up bias. *Journal of experimental social psychology* 74, 187–195.
- Allaire, J. J., Y. Xie, C. Dervieux, J. McPherson, J. Luraschi, K. Ushey, A. Atkins, H. Wickham, J. Cheng, W. Chang, and R. Iannone (2024). *rmarkdown: Dynamic Documents for R*. R package version 2.29.
- Almeida, L. N., M. Behlau, N. dos Santos Ramos, I. K. Barbosa, and A. A. Almeida (2022). Factor analysis of the brazilian version of the voice-related quality of life (v-rqol) questionnaire. *Journal of Voice* 36(5), 736.e17–736.e24.
- Angrist, J. D. and J.-S. Pischke (2015). *Mastering 'Metrics: The Path From Cause to Effect*. Princeton: Princeton University Press.
- Armstrong, J. S. and P. Soelberg (1968). On the interpretation of factor analysis. *Psychological Bulletin* 70(5), 361–364.
- Ashton, M. C. and K. Lee (2009). The hexaco-60: A short measure of the major dimensions of personality. *Journal of Personality Assessment* 91(4), 340–345.
- Barnard, J., R. McCulloch, and X.-L. Meng (2000). Modeling covariance matrices in terms of standard deviations and correlations, with application to shrinkage. *Statistica Sinica* 10(4), 1281–1311.
- Bartlett, M. S. (1950). Tests of significance in factor analysis. *British Journal of Psychology* 3(2), 77–85.
- Bayes, T. and R. Price (1763). An essay towards solving a problem in the doctrine of chance. by the late Rev. Mr. Bayes, communicated by Mr. Price, in a letter to John Canton, A.M.F.R.S. *Philosophical Transactions of the Royal Society of London* 53, 370–417.
- Ben-Shachar, M. S., D. Lüdtke, and D. Makowski (2020). effectsize: Estimation of effect size indices and standardized parameters. *Journal of Open Source Software* 5(56), 2815.
- Berger, J. O. and L. R. Pericchi (2001). Objective bayesian methods for model selection: Introduction and comparison. *IMS Lecture notes—Monograph series, Model selection* 38, 135–207.
- Bolch, B. W. (1968). More on unbiased estimation of the standard deviation. *The American Statistician* 22(3), 27.
- Boyd, S. P. and L. Vandenberghe (2018). *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* (1 ed.). Cambridge University Press.
- Brachet, P. (2025). *Texmaker*. Version v6.0.1.
- Bravais, A. (1844). *Analyse mathématique sur les probabilités des erreurs de situation d'un point*. Impr. royale. In French.
- Brock, K. (2023). *trialr: Clinical Trial Designs in 'rstan'*. R package version 0.1.6.
- Brooks, S., A. Gelman, G. Jones, and X.-L. Meng (2011). *Handbook of Markov Chain Monte Carlo* (1 ed.). Chapman and Hall.

- Brown, T. A. (2015). *Confirmatory factor analysis for applied research* (2 ed.). Guilford.
- Bunyakovsky, V. (1859). Sur quelques inegalités concernant les intégrales aux différences finies. *Mémoires présentés à l'Académie impériale des sciences de St. Petersbourg* 7(1), 6.
- Burgess, J. P. (2022[1948]). *Set Theory*. Cambridge University Press.
- Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software* 80(1), 1–28.
- Bürkner, P.-C. (2018). Advanced Bayesian multilevel modeling with the R package brms. *The R Journal* 10(1), 395–411.
- Cajori, F. (2007). *A history of mathematical notations* (2 ed.). Cosimo Classics.
- Cantor, G. (1874). Über eine eigenschaft des inbegriffs aller reellen algebraischen zahlen. *Journal für die Reine und Angewandte Mathematik* 77, 258–262. In German.
- Carroll, R. M. and L. A. Nordholm (1975). Sampling characteristics of Kelley's epsilon and Hays' omega. *Educational and Psychological Measurement* 35(3), 541–554.
- Casella, G. and R. L. Berger (2002). *Statistical Inference* (2 ed.). Duxbury.
- Cauchy, A.-L. (1821). Sur les formules qui résultent de l'emploi du signe $>$ ou $<$, et sur les moyennes entre plusieurs quantités. *Cours d'Analyse, 1er Partie: Analyse Algébrique. OEuvres Ser.2 III*, 373–377.
- Champely, S. (2020). *pwr: Basic Functions for Power Analysis*. R package version 1.3-0.
- Clapham, C. and J. Nicholson (2014). *Oxford Concise Dictionary of Mathematics* (5 ed.). Oxford University Press.
- Clayton, A. (2021). *Bernoulli's Fallacy: Statistical Illogic and the Crisis of Modern Science*. Columbia University.
- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2 ed.). Lawrence Erlbaum Associates.
- Cohen, J. (1991). A power primer. *Psychological Bulletin* 112(1), 155–159.
- Cohen, J. (1994). The earth is round ($p < .05$). *The American Psychologist* 49(21), 997–1003.
- Comrey, A. L. and H. B. Lee (1992). *A first course in factor analysis* (2 ed.). Erlbaum.
- Coombes, K. R., G. Brock, Z. B. Abrams, and L. V. Abruzzo (2019). Polychrome: Creating and assessing qualitative palettes with many colors. *Journal of Statistical Software, Code Snippets* 90(1), 1–23.
- Danaei, Z., F. Madadzadeh, F. Sheikhshoei, and H. Dehdarirad (2024). Translation and cross-cultural adaptation of persian version of evidence based medicine questionnaire (ebmq) in postgraduate medical students in iran. *PLoS One* 19(4), e0301831.
- de Finetti, B. (1974). *Theory of probability, Volume I*. Wiley.
- Descartes, R. (1886[1637]). *La Géométrie*. Paris A. Hermann. In French.

- Downes, M., L. C. Gurrin, D. R. English, J. Pirkis, D. Currier, M. J. Spittal, and J. B. Carlin (2018). Multilevel regression and poststratification: A modeling approach to estimating population quantities from highly selected survey samples. *American Journal of Epidemiology* 187(8), 1780–1790.
- Dziuban, C. D. and E. C. Shirkey (1974). When is a correlation matrix appropriate for factor analysis? some decision rules. *Psychological Bulletin* 81(6), 358–361.
- Eddelbuettel, D. (2013). *Seamless R and C++ Integration with Rcpp*. New York: Springer. ISBN 978-1-4614-6867-7.
- Eddelbuettel, D. (2017). *random: True Random Numbers using RANDOM.ORG*. R package version 0.2.6.
- Eddelbuettel, D. and J. J. Balamuta (2018). Extending R with C++: A Brief Introduction to Rcpp. *The American Statistician* 72(1), 28–36.
- Eddelbuettel, D. and R. François (2011). Rcpp: Seamless R and C++ integration. *Journal of Statistical Software* 40(8), 1–18.
- Eddelbuettel, D., R. François, J. J. Allaire, K. Ushey, Q. Kou, N. Russell, I. Ucar, D. Bates, and J. Chambers (2025). *Rcpp: Seamless R and C++ Integration*. R package version 1.1.0.
- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics* 7(1), 1–26.
- Efron, B. (1987). Better bootstrap confidence intervals. *Journal of the American Statistical Association* 82(397), 171–185.
- Efron, B. (2003). Second thoughts on the bootstrap. *Statistical Science* 18(2), 135–140.
- Efron, B. and R. Tibshirani (1994). *An introduction to the bootstrap*. Chapman and Hall.
- Euler, L. (1755). *Institutiones calculi differentialis*. Academia Imperialis Scientiarum. In Latin.
- Field, A. (2018). *Discovering Statistics Using IBM SPSS Statistics* (5 ed.). SAGE.
- Funder, D. C. and D. J. Ozer (2019). Evaluating effect size in psychological research: sense and nonsense. *Advances in Methods and Practices in Psychological Science* 2(2), 155–168.
- Gabry, J., D. Simpson, A. Vehtari, M. Betancourt, and A. Gelman (2019). Visualization in bayesian workflow. *The Journal of the Royal Statistical Society, Series A (Statistics in Society)* 182, 389–402.
- Gauss, C. F. (2012[1809]). *Theoria Motus Corporum Coelestium In Sectionibus Conicis Solem Ambientium*. Cambridge University Press. In Latin. Edited by Ernst Julius Schering.
- Gelman, A. (2022). The development of bayesian statistics. *Journal of the Indian Institute of Science* 102(4), 1131–1134.
- Gelman, A. (2023). What is a standard error? *Journal of Econometrics* 237(1), 105516.
- Gelman, A. and J. B. Carlin (2014). Beyond power calculations: Assessing type S (sign) and type M (magnitude) errors. *Perspectives on Psychological Science* 9(6), 641–651.

- Gelman, A., J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, and D. B. Rubin (2014). *Bayesian Data Analysis* (3 ed.). Chapman and Hall.
- Gelman, A., B. Goodrich, J. Gabry, and A. Vehtari (2019). R-squared for bayesian regression models. *The American Statistician* 73(3), 307–309.
- Gelman, A. and J. Hill (2007). *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press.
- Gelman, A., J. Hill, and A. Vehtari (2021). *Regression and Other Stories*. Cambridge University Press.
- Gelman, A., X.-L. Meng, and H. Stern (1996). Posterior predictive assessment of model fitness via realized discrepancies. *Statistica Sinica* 6(4), 733–760.
- Gelman, A. and D. B. Rubin (1992). Inference from iterative simulation using multiple sequences. *Statistical Science* 7(4), 457–472.
- Gelman, A. and H. Stern (2006). The difference between “significant” and “not significant” is not itself statistically significant. *The American Statistician* 60(4), 328–331.
- Ghadiri, M., M. Tavakoli, and S. Ketabi (2015). Development of a criticality-oriented english teaching perceptions inventory (cetpi) and exploring its internal consistency and underlying factor structure. *Current Psychology (New Brunswick, N.J.)* 34(2), 268–281.
- Gigerenzer, G. (2004). Mindless statistics. *The Journal of Socio-Economics* 33(5), 587–606.
- Gigerenzer, G. (2018). Statistical rituals: The replication delusion and how we got there. *Advances in Methods and Practices in Psychological Science* 1(2), 198–218.
- Gignac, G. E. and E. T. Szodorai (2016). Effect size guidelines for individual differences researchers. *Personality and individual differences* 102, 74–78.
- Gosset, W. S. (1908). The probable error of a mean. *Biometrika* 6(1), 1–25.
- Grolemund, G. and H. Wickham (2011). Dates and times made easy with lubridate. *Journal of Statistical Software* 40(3), 1–25.
- Gruber, J. (2014). *The Markdown File Extension*.
- Gumbel, E. J. (1933). Spurious correlation and its significance to physiology. *Mathematical Proceedings of the Cambridge Philosophical Society* 21(154), 179–194.
- Harman, H. H. (1967). *Modern factor analysis*. University of Chicago Press.
- Harrell Jr, F. E. (2025). *Hmisc: Harrell Miscellaneous*. R package version 5.2-3.
- Hastie, T. J., R. J. Tibshirani, and J. Friedman (2017). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2 ed.). Springer.

- Hays, W. (1963). *Statistics for Psychologists*. Holt, Rinehart and Winston.
- Helmert, F. R. (1875). Über die berechnung des wahrscheinlichen fehlers aus einer endlichen anzahl wahrer beobachtungsfehler. *Zeitschrift für Angewandte Mathematik und Physik* 20, 300–303. In German.
- Helmert, F. R. (1876a). Über die wahrscheinlichkeit der potenzsummen der beobachtungsfehler und über einige damit in zusammenhang stehende fragen. *Zeitschrift für Angewandte Mathematik und Physik* 21, 192–218. In German.
- Helmert, F. R. (1876b). Die genauigkeit der formel von peters zur berechnung des wahrscheinlichen beobachtungsfehlers directer beobachtungen gleicher genauigkeit. *Astronomische Nachrichten* 88(8-9), 113–132. In German.
- Hoffman, M. D. and A. Gelman (2014). The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research* 15, 1593–1623.
- Howell, D. C. (2012). *Statistical methods for psychology* (8 ed.). Wadsworth.
- IBM (2025). *IBM SPSS Statistics: KMO and Bartlett's Test*. IBM. Accessed 10/4/2025.
- Imai, K. (2017). *Quantitative Social Science: An Introduction*. Princeton University Press.
- Jeffreys, H. (1946). An invariant form for the prior probability in estimation problems. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences* 186(1007), 453–461.
- Jensen, J. L. W. V. (1906). Sur les fonctions convexes et les inégalités entre les valeurs moyennes. *Acta Mathematica* 30, 175–193. In French.
- Jung, L. (2024). *scrutiny: Error Detection in Science*. R package version 0.5.0.
- Juárez, M. A. and M. F. J. Steel (2010). Model-based clustering of non-gaussian panel data based on skew-t distributions. *Journal of Business and Economic Statistics* 28(1), 52–66.
- Kaiser, H. F. (1970). A second generation little jiffy. *Psychometrika* 35(4), 401–415.
- Kaiser, H. F. (1974). An index of factor simplicity. *Psychometrika* 39(1), 31–36.
- Kaiser, H. F. (1981). A revised measure of sampling adequacy for factor-analytic data matrices. *Educational and Psychological Measurement* 41(2), 379–381.
- Kaiser, H. F. and J. Rice (1974). Little jiffy, mark iv. *Educational and Psychological Measurement* 34(1), 111–117.
- Kass, R. E., B. P. Carlin, A. Gelman, and R. M. Neal (1998). Markov chain Monte Carlo in practice: A roundtable discussion. *The American Statistician* 52(2), 93–100.
- Kay, M. (2024a). ggdist: Visualizations of distributions and uncertainty in the grammar of graphics. *IEEE Transactions on Visualization and Computer Graphics* 30(1), 414–424.
- Kay, M. (2024b). *ggdist: Visualizations of Distributions and Uncertainty*. R package version 3.3.2.

- Kay, M. (2024c). *tidybayes: Tidy Data and Geoms for Bayesian Models*. R package version 3.0.7.
- Kelley, T. L. (1935). An unbiased correlation ratio measure. *Proceedings of the National Academy of Sciences* 21(9), 554–559.
- Kendall, M. G. (1938). A new measure of rank correlation. *Biometrika* 30(1-2), 81–89.
- King, G. (1998). *Unifying Political Methodology - The Likelihood Theory of Statistical Inference*. The University Of Michigan Press.
- Kruschke, J. K. (2014). *Doing Bayesian data analysis: A tutorial with R, JAGS, and Stan* (2 ed.). Academic Press.
- Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science* 8(4), 355–363.
- Lamport, L. (1986). *LATEX: a document preparation system*. Addison-Wesley Pub. Co.
- Laplace, P. S. (2009[1814]). *Essai philosophique sur les probabilités* (5 ed.). Cambridge: Cambridge University Press. In French.
- Lee, M. D. and M. R. Webb (2005). Modeling individual differences in cognition. *Psychonomic Bulletin and Review* 12(4), 605–621.
- Legendre, A. M. (1805). *Nouvelles méthodes pour la détermination des orbites des comètes*. F. Didot. In French.
- Lenz, G. S. and A. Sahn (2021). Achieving statistical significance with covariates and without transparency. *Political Analysis* 29(3), 356–369.
- Levy, R. and R. J. Mislevy (2020). *Bayesian Psychometric Modeling*. Chapman and Hall.
- Lewandowski, D., D. Kurowicka, and H. Joe (2009). Generating random correlation matrices based on vines and extended onion method. *Journal of Multivariate Analysis* 100(9), 1989–2001.
- Li, R. and S. Nadarajah (2020). A review of student's t distribution and its generalizations. *Empirical Economics* 58(3), 1461–1490.
- Lindley, D. V. (1972). *Bayesian statistics, a review*. SIAM.
- Lovakov, A. and E. R. Agadullina (2021). Empirically derived guidelines for effect size interpretation in social psychology. *European Journal of Social Psychology* 51(3), 485–504.
- Lüdtke, D., M. S. Ben-Shachar, I. Patil, P. Waggoner, and D. Makowski (2021). performance: An R package for assessment, comparison and testing of statistical models. *Journal of Open Source Software* 6(60), 3139.
- Lüroth, J. (1876). Vergleichung von zwei werten des wahrscheinlichen fehlers. *Astronomische Nachrichten* 87(14), 209–220. In German.
- Makowski, D., M. S. Ben-Shachar, S. H. A. Chen, and D. Lüdtke (2019a). Indices of effect existence and significance in the bayesian framework. *Frontiers in Psychology* 10, 2767.

- Makowski, D., M. S. Ben-Shachar, and D. Lüdtke (2019b). bayestestr: Describing effects and their uncertainty, existence and significance within the bayesian framework. *Journal of Open Source Software* 4(4), 1541.
- McElreath, R. (2019). *Statistical Rethinking: A Bayesian Course with Examples in R* (2 ed.). Chapman and Hall.
- Meschiari, S. (2022). *latex2exp: Use LaTeX Expressions in Plots*. R package version 0.9.6.
- Microsoft (2024). *Windows 11*. x64, build 26100.
- Microsoft and S. Weston (2022a). *doParallel: Foreach Parallel Adaptor for the 'parallel' Package*. R package version 1.0.17.
- Microsoft and S. Weston (2022b). *foreach: Provides Foreach Looping Construct*. R package version 1.5.2.
- Moran, P. A. P. (1948). Rank correlation and product-moment correlation. *Biometrika* 35(1/2), 203–206.
- Mordkoff, J. T. (2019). A simple method for removing bias from a popular measure of standardized effect size: Adjusted partial eta squared. *Advances in Methods and Practices in Psychological Science* 2(3), 228–232.
- Müller, K. and H. Wickham (2025). *tibble: Simple Data Frames*. R package version 3.3.0.
- Nahin, P. (1998). *An Imaginary Tale: the Story of the Square Root of -1*. Princeton University Press.
- Neyman, J. and E. S. Pearson (1933). The testing of statistical hypotheses in relation to probabilities a priori. *Mathematical Proceedings of the Cambridge Philosophical Society* 29(4), 492–510.
- O'Hagan, A. (1979). On outlier rejection phenomena in bayes inference. *Journal of the Royal Statistical Society. Series B (Methodological)* 41(3), 358–367.
- Park, C., H. Kim, and M. Wang (2022). Investigation of finite-sample properties of robust location and scale estimators. *Communications in Statistics - Simulation and Computation* 51(5), 2619–2645.
- Park, C. and M. Wang (2020). A study on the x-bar and s control charts with unequal sample sizes. *Mathematics* 8(5), 698.
- Park, C. and M. Wang (2022). *rQCC: Robust Quality Control Chart*. R package version 2.22.12.
- Pasek, J., A. Tahk, G. Culter, and M. Schwemmler (2025). *weights: Weighting and Weighted Statistics*. R package version 1.1.2.
- Patil, I., D. Makowski, M. S. Ben-Shachar, B. M. Wiernik, E. Bacher, and D. Lüdtke (2022). datawizard: An R package for easy data preparation and statistical transformations. *Journal of Open Source Software* 7(78), 4684.
- Pearson, K. (1895a). Notes on regression and inheritance in the case of two parents. *Proceedings of the Royal Society of London* 58(347), 240–242.
- Pearson, K. (1895b). Contributions to the mathematical theory of evolution. ii. skew variation in homogeneous material. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 186(374), 343–414.

- Posit Team (2025). *RStudio: Integrated Development Environment for R*. Boston, MA: Posit Software, PBC.
- R Core Team (2025). *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing.
- Revelle, W. (2025). *psych: Procedures for Psychological, Psychometric, and Personality Research*. Evanston, Illinois: Northwestern University. R package version 2.5.6.
- Revolution Analytics and S. Weston (2022). *iterators: Provides Iterator Construct*. R package version 1.0.14.
- Rinker, T. W. and D. Kurkiewicz (2018). *pacman: Package Management for R*. Buffalo, New York. version 0.5.0.
- Robert, C. and G. Casella (2013). *Monte Carlo Statistical Methods* (2 ed.). Springer Nature.
- Rubinstein, R. Y. and D. P. Kroese (2016). *Simulation and the Monte Carlo method* (3 ed.). Wiley.
- Rudolff, C. (1615[1525]). *Die Coss*. Wilhelm Janson.
- Sadowska, A., M. Nowak, and B. Czarkowska-Pączek (2021). Assessment of the reliability of the Polish language version of the fatcod-b scale among nursing students. *Journal of Cancer Education* 36(3), 561–566.
- Sarfraz, S., A. Surti, R. Ali, R. Rehman, A. Heboyan, and N. Ahmed (2024). Faculty application and perceived effectiveness of cognitive psychology principles in medical education. a mixed method study. *BMC Medical Education* 24(1), 911.
- Schwarz, H. A. (1888). Über ein flächen kleinsten flächeninhalts betreffendes problem der variationsrechnung. *Acta Societatis Scientiarum Fennicae XV*, 318.
- Schäfer, J. and K. Strimmer (2005). A shrinkage approach to large-scale covariance estimation and implications for functional genomics. *Statistical Applications in Genetics and Molecular Biology* 4(1), 32.
- Sen, P. K. and J. M. Singer (1993). *Large Sample Methods in Statistics: An Introduction with Applications*. Chapman and Hall.
- Sharpsteen, C. and C. Bracken (2023). *tikzDevice: R Graphics Output in LaTeX Format*. R package version 0.12.6.
- Shirkey, E. C. and C. D. Dzuban (1976). A note on some sampling characteristics of the measure of sampling adequacy (msa). *Multivariate Behavioral Research* 11(1), 125–128.
- Simpson, D., H. Rue, A. Riebler, T. G. Martins, and S. H. Sørbye (2017). Penalising model component complexity: A principled, practical approach to constructing priors. *Statistical Science* 32(1), 1–28.
- Spearman, C. (1904). The proof and measurement of association between two things. *The American Journal of Psychology* 15(1), 72–101.
- Spearman, C. (1910). Correlation calculated with faulty data. *British Journal of Psychology* 3(3), 271–295.
- Stan Development Team (2020). *StanHeaders: Headers for the R interface to Stan*. R package version 2.23.10.
- Stan Development Team (2025a). *Stan Reference Manual*. Version 2.37.0 (2 September 2025).

- Stan Development Team (2025b). RStan: the R interface to Stan. R package version 2.32.7.
- Stewart, J. (2010). *Calculus: Early Transcendentals* (7 ed.). Cengage Learning.
- Stigler, S. M. (1989). Francis Galton's account of the invention of correlation. *Statistical Science* 4(2), 73–79.
- Stock, J. and M. W. Watson (2019). *Introduction to Econometrics* (4 ed.). Pearson Education Limited. Global Edition.
- Stone, M. (1974). Cross-validated choice and assessment of statistical predictions. *Journal of the Royal Statistical Society, Series B (Methodological)* 36(2), 111–147.
- Strahan, R. S. (1982). Assessing magnitude of effect from rank-order correlation coefficients. *Educational and Psychological Measurement* 42(3), 763–765.
- The LaTeX Project (2025). *The LaTeX Project*. The LaTeX Project.
- Thành, H. T. and D. E. Knuth (2025). *MikTeX*. MiKTeX-pdfTeX version 4.21 (MiKTeX version 25.4).
- Ushey, K., J. J. Allaire, H. Wickham, and G. Ritchie (2024). *rstudioapi: Safely Access the RStudio API*. R package version 0.17.1.
- van Buuren, S. and K. Groothuis-Oudshoorn (2011). mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software* 45(3), 1–67.
- Vehtari, A., A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of mcmc (with discussion). *Bayesian Analysis* 16(2), 667–718.
- Wagenmakers, E.-J., T. Lodewyckx, H. Kuriyal, and R. Grasman (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage-Dickey method. *Cognitive Psychology* 60(3), 158–189.
- Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York.
- Wickham, H. (2023a). *forcats: Tools for Working with Categorical Variables (Factors)*. R package version 1.0.0.
- Wickham, H. (2023b). *stringr: Simple, Consistent Wrappers for Common String Operations*. R package version 1.5.1.
- Wickham, H., M. Averick, J. Bryan, W. Chang, L. D. McGowan, R. François, G. Golemund, A. Hayes, L. Henry, J. Hester, M. Kuhn, T. L. Pedersen, E. Miller, S. M. Bache, K. Müller, J. Ooms, D. Robinson, D. P. Seidel, V. Spinu, K. Takahashi, D. Vaughan, C. Wilke, K. Woo, and H. Yutani (2019). Welcome to the tidyverse. *Journal of Open Source Software* 4(43), 1686.
- Wickham, H., R. François, L. Henry, K. Müller, and D. Vaughan (2023). *dplyr: A Grammar of Data Manipulation*. R package version 1.1.4.
- Wickham, H. and L. Henry (2025). *purrr: Functional Programming Tools*. R package version 1.1.0.

- Wickham, H., J. Hester, and J. Bryan (2024a). *readr: Read Rectangular Text Data*. R package version 2.1.5.
- Wickham, H., D. Vaughan, and M. Girlich (2024b). *tidyr: Tidy Messy Data*. R package version 1.3.1.
- Wooldridge, J. M. (2019). *Introductory Econometrics* (7 ed.). Cengage.
- Xie, Y. (2014). knitr: A comprehensive tool for reproducible research in R. In V. Stodden, F. Leisch, and R. D. Peng (Eds.), *Implementing Reproducible Computational Research*. Chapman and Hall. ISBN 978-1466561595.
- Xie, Y. (2015). *Dynamic Documents with R and knitr* (2 ed.). Boca Raton, Florida: Chapman and Hall. ISBN 978-1498716963.
- Xie, Y. (2016). *bookdown: Authoring Books and Technical Documents with R Markdown*. Boca Raton, Florida: Chapman and Hall.
- Xie, Y. (2025a). *bookdown: Authoring Books and Technical Documents with R Markdown*. R package version 0.43.
- Xie, Y. (2025b). *knitr: A General-Purpose Package for Dynamic Report Generation in R*. R package version 1.5.
- Xie, Y., J. Allaire, and G. Golemund (2018). *R Markdown: The Definitive Guide*. Boca Raton, Florida: Chapman and Hall.
- Xie, Y., C. Dervieux, and E. Riederer (2020). *R Markdown Cookbook*. Boca Raton, Florida: Chapman and Hall.
- Zabihi, R., S. Ketabi, M. Tavakoli, and M. Ghadiri (2014). Examining the internal consistency reliability and construct validity of the authentic happiness inventory (ahi) among iranian efl learners. *Current Psychology (New Brunswick, N.J.)* 33(3), 377–392.
- Zhu, H. (2024). *kableExtra: Construct Complex Table with 'kable' and Pipe Syntax*. R package version 1.4.0.
- Zhu, M. and A. Y. Lu (2004). The counter-intuitive non-informative prior for the bernoulli family. *Journal of Statistics Education* 12(2).
- Zuur, A. F., E. N. Ieno, and C. S. Elphick (2010). A protocol for data exploration to avoid common statistical problems: Data exploration. *Methods in Ecology and Evolution* 1(1), 3–14.

6 Appendix

This appendix contains the specifications for software used in this paper (Section 6.1); the R function implementing the Bayesian Kaiser-Meyer-Olkin (BKMO) index (Section 6.2); the formula and R function implementations of the different versions of the frequentist KMO index (Section 6.3); the results from the Monte Carlo simulations assessing the characteristics of the frequentist KMO index discussed in section 2.3 (Section 6.4); and a results matrix containing the frequentist inter-correlations, anti-image inter-correlations, and KMO indices of the simulated data used in section 2.4 (Section 6.5).

Besides this appendix, supplementary materials, including data and code, are made freely available online on the *open science framework* (OSF) project page dedicated to this paper (DOI: 10.17605/OSF.IO/T3UPD).

6.1 Software specifications

This document was written with *markdown* (Gruber, 2014) and $\text{\LaTeX}$ (Lamport, 1986; The LaTeX Project, 2025), using $\text{\pdfTeX}$ (v3.141592653-2.6-1.40.2; $\text{\MiKTeX}$ [v25.4], Thành and Knuth, 2025) and the TeXmaker IDE (v6.0.1, Brachet, 2025), compiled on a Windows 11 x64 (build 26100) operating system (OS, Microsoft, 2024) on the 2025-10-16 16:02 CEST (Timezone: Europe/Copenhagen). The machine has 16 cores available, and the range of doubles is $2.23 \times 10^{-308} - 1.8 \times 10^{+308}$.

All simulations and analyses were conducted on the aforementioned soft-/hardware using the R programming language (v4.4.3 [2025-02-28 ucrt], R Core Team, 2025) in the RStudio IDE (2025.5.1.513, Posit Team, 2025), using the R packages *bayesplot* (v1.13.0, Gabry et al., 2019), *bayestestR* (v0.16.1, Makowski et al., 019b), *bookdown* (v0.43, Xie, 2016, 025a), *brms* (v2.22.0, Bürkner, 2017, 2018), *datawizard* (v1.1.0, Patil et al., 2022), *doParallel* (v1.0.17, Microsoft and Weston, 022a), *dplyr* (v1.1.4, Wickham et al., 2023), *effectsize* (v1.0.1, Ben-Shachar et al., 2020), *forcats* (v1.0.0, Wickham, 023a), *foreach* (v1.5.2, Microsoft and Weston, 022b), *ggplot2* (v3.5.2, Wickham, 2016), *Hmisc* (v5.2.3, Harrell Jr, 2025), *iterators* (v1.0.14, Revolution Analytics and Weston, 2022), *kableExtra* (v1.4.0, Zhu, 2024), *knitr* (v1.50, Xie, 2014, 2015, 025b), *latex2exp* (v0.9.6, Meschiari, 2022), *lubridate* (v1.9.4, Grolemund and Wickham, 2011), *pacman* (v0.5.1, Rinker and Kurkiewicz, 2018), *parallel* (v4.4.3, R Core Team, 2025), *performance* (v0.15.0, Lüdtke et al., 2021), *Polychrome* (v1.5.4, Coombes et al., 2019), *psych* (v2.5.6, Revelle, 2025), *purrr* (v1.1.0, Wickham and Henry, 2025), *pwr* (v1.3.0, Champely, 2020), *Rcpp* (v1.1.0, Eddebuettel, 2013; Eddebuettel et al., 2025; Eddebuettel and Balamuta, 2018; Eddebuettel and François, 2011), *readr* (v2.1.5, Wickham et al., 024a), *rmarkdown* (v2.29, Allaire et al., 2024; Xie et al., 2018, 2020), *rQCC* (v2.22.12, Park et al., 2022; Park and Wang, 2020, 2022), *rstan* (v2.32.7, Stan Development Team, 025b), *rstudioapi* (v0.17.1, Ushey et al., 2024), *scrutiny* (v0.5.0, Jung, 2024), *StanHeaders* (v2.32.10, Stan Development Team, 2020), *stringr* (v1.5.1, Wickham, 023b), *tibble* (v3.3.0, Müller and Wickham, 2025), *tidybayes* (v3.0.7, Kay, 024c), *tidyr* (v1.3.1, Wickham et al., 024b), *tidyverse* (v2.0.0, Wickham et al., 2019), *tikzDevice* (v0.12.6, Sharpsteen and Bracken, 2023), *trialr* (v0.1.6, Brock, 2023), and *weights* (v1.1.2, Pasek et al., 2025). The seed used for generating random numbers (e.g., for MCMC) was exogenously predetermined using the *random* R package (v0.2.6, Eddebuettel, 2017).

6.2 BKMO function

A simple implementation of a function for the *Bayesian Kaiser-Meyer-Olkin* (BKMO) index in R (v4.4.3 [2025-02-28 ucrt], R Core Team, 2025), which computes the posterior ‘sampling adequacy’ from the posterior inter-correlations estimated by the *brms* R package (v2.22.0, Bürkner, 2017, 2018), is provided in the code below:

```
1 # Compute Bayesian KMO index from a Bayesian multivariate linear model
2 #
3 # @param r A two-dimensional matrix, where rows are posterior draws and columns are
4 #   ↳ manifestations, containing the estimated inter-correlations from a Bayesian
5 #   ↳ multivariate linear model
6 # @return A matrix of Bayesian KMO indices, where rows are posterior draws (with
7 #   ↳ the number of rows equal to the number of posterior draws) and columns are
8 #   ↳ the individual BKMO indices and one overall BKMO index (with the number of
```

↪ columns equal to the number of theorized manifestations plus one).

```

5
6 BKMO <- function(r = NULL){
7   if(!is.matrix(r)) r <- as.matrix(r)
8
9   # Compute p (i.e., number of theorized manifestations)
10  disc <- sqrt(1 + 8 * ncol(r))
11  p <- (1 + disc) / 2
12
13  # Define utility function for vectorized operation
14  utility_KMO <- function(r = r, p = p){
15    # Construct correlation matrix
16    R <- diag(1, p)
17    R[upper.tri(R)] <- r
18    R[lower.tri(R)] <- t(R)[lower.tri(R)]
19    row.names(R) <- paste0('m', 1:p)
20    colnames(R) <- paste0('m', 1:p)
21
22    # Compute inverse R
23    R_inv <- solve(R)
24
25    # Compute anti-image R
26    R_q <- -R_inv / sqrt(outer(diag(R_inv), diag(R_inv)))
27
28    # Compute KMO
29    r2 <- R^2
30    q2 <- R_q^2
31    diag(r2) <- 0
32    diag(q2) <- 0
33    kmo <- c(
34      rowSums(r2) / (rowSums(r2) + rowSums(q2)),
35      overall = sum(r2) / (sum(r2) + sum(q2))
36    )
37    return(kmo)
38  }
39
40  # Compute KMO for each posterior draw using vectorization
41  bkmo <- do.call(rbind, lapply(1:nrow(r), function(i) utility_KMO(
42    r = r[i, , drop = FALSE], p = p
43  )
44  )
45  )

```

```
46   return(as.data.frame(bkmo))
47 }

1  ## Example
2
3  # Simulate data
4  library("tidyverse")
5  sample_size <- 1000
6  n_manifestations <- 10
7  sesoi <- .3
8  loadings <- runif(n_manifestations, sesoi, .7)
9  df <- data.frame(
10   latent = rnorm(sample_size)
11 )
12 for(i in 1:n_manifestations){
13   df[,paste0("m", i)] <- loadings[i]*df$latent + rnorm(sample_size)
14 }
15 df <- df %>% select(-latent)
16
17 # Define maximum-likelihood functions
18 sd2 <- function(x = NULL, na.rm = FALSE){
19   stopifnot(is.numeric(x))
20   stopifnot(is.logical(na.rm))
21
22   x_mean <- mean(x, na.rm = na.rm)
23   x_var <- mean((x - x_mean)^2, na.rm = na.rm)
24   x_sd <- sqrt(x_var)
25   return(x_sd)
26 }
27 standardize2 <- function(x = NULL, na.rm = FALSE){
28   stopifnot(is.numeric(x))
29   stopifnot(is.logical(na.rm))
30
31   x_mean <- mean(x, na.rm = na.rm)
32   x_var <- mean((x - x_mean)^2, na.rm = na.rm)
33   x_sd <- sqrt(x_var)
34
35   x_standardized <- (x - x_mean) / x_sd
36   return(x_standardized)
37 }
38
39 # Standardize data matrix
40 df <- apply(df, 2, standardize2) %>% as.data.frame()
```

```
41 |
42 | # Specify MCMC
43 | library("parallel")
44 | posterior_samples <- 4000
45 | chains <- cores <- parallel::detectCores()-1
46 | warmup <- 1000
47 | iter <- ceiling((posterior_samples / chains) + warmup)
48 | posterior_samples <- (iter - warmup)*chains
49 |
50 | # Specify Bayesian Generalized Linear Model (BGLM)
51 | library("brms")
52 | bglm_model <- brms::bf(
53 |   mvbind(m1, m2, m3, m4, m5, m6, m7, m8, m9, m10) ~ 0,
54 |   sigma ~ 0,
55 |   family = gaussian(
56 |     link = "identity"
57 |   )
58 | ) + set_rescor(TRUE)
59 |
60 |
61 | # Specify 'weakly-informative' model priors
62 | bglm_priors <- brms::get_prior(bglm_model, data = df)
63 | bglm_priors[1,] <- brms::set_prior(
64 |   "lkj(2)",
65 |   class = "rescor"
66 | )
67 |
68 | # Fit Bayesian multivariate linear model with brms
69 | bglm_fit <- brms::brm(
70 |   formula = bglm_model,
71 |   family = gaussian(
72 |     link = "identity"
73 |   ),
74 |   prior = bglm_priors,
75 |   data = df,
76 |   chains = chains,
77 |   cores = cores,
78 |   iter = iter,
79 |   warmup = warmup
80 | )
81 |
82 | # Extract posterior draws of the (residual) correlations from the fitted Bayesian
```

```

      ↪ model
83 library("stringr")
84 bglm_samples <- brms::as_draws_df(bglm_fit)
85 bglm_correlations <- bglm_samples[, stringr::str_detect(colnames(bglm_samples), "
      ↪ rescor_")]
86
87 # Compute Bayesian KMO index
88 library("bayestestR")
89 kmo_bayes <- BKMO(r = bglm_correlations)
90 hist(kmo_bayes$overall)
91 print(
92   paste0(
93     "Mean = ", mean(kmo_bayes$overall) %>% round(digits = 3),
94     " (SD = ", sd2(kmo_bayes$overall) %>% round(digits = 3),
95     "; 95% HDI[",
96     bayestestR::hdi(kmo_bayes$overall)$CI_low %>% round(digits = 3), "; ",
97     bayestestR::hdi(kmo_bayes$overall)$CI_high %>% round(digits = 3), "]" )
98 )
99 )

```

Generalizing the code for more sophisticated scenarios is trivial but beyond the scope of this paper. As discussed in section 3.5, researchers interested in robust versions of the BKMO can simply rank-standardize their variables before computing the posterior correlation matrix. Alternatively, they can change the multivariate likelihood from a Gaussian to a Student t by manually editing the Stan code produced by brms (Bürkner, 2017, 2018), since a Student t multivariate model is currently unsupported by that package.

6.3 Measures of sampling adequacy

The formula underlying the *Kaiser-Meyer-Olkin* (KMO) indices discussed in section 2.3, more formally denoted *measures of sampling adequacy* (MSA), are provided here. These formula are based on the notation used throughout this paper, relying on the frequentist conceptualizations (see Section 2.2). This section also includes code implementing each of these MSAs as functions in R. Section 6.3.1 covers the 'Mark II' version (i.e., Kaiser, 1970), section 6.3.2 covers 'Mark IV' (i.e., Kaiser and Rice, 1974), and section 6.3.3 covers 'Mark V' (i.e., Kaiser, 1981).

6.3.1 Kaiser (1970)

The initial MSA (i.e., MSA_1), also known as 'second generation Little Jiffy', proposed by (Kaiser, 1970). The formula for calculating the overall MSA is provided in definition (7.4.1.1).

$$MSA_1 \equiv 1 - \frac{\sum_{i \neq j} \sum \hat{\rho}_{q[i,j]}^2}{\sum_{i \neq j} \sum \hat{\rho}_{i,j}^2} \quad (7.4.1.1)$$

where $\sum \sum$ denotes the row-wise summation operator, $\hat{\rho}_q$ denotes the estimated anti-image product-moment correlation coefficient, and $\hat{\rho}$ denotes the estimated Pearson product-moment correlation coefficient.

The formula for calculating individual MSAs are provided in definition (7.4.1.2).

$$\text{MSA}_1(i) \equiv 1 - \frac{\sum_{j,j \neq i} \hat{\rho}_{q[i,j]}^2}{\sum_{j,j \neq i} \hat{\rho}_{i,j}^2} \quad (7.4.1.2)$$

where Σ denotes the summation operator, $\hat{\rho}_q$ denotes the estimated anti-image product-moment correlation coefficient, and $\hat{\rho}$ denotes the estimated Pearson product-moment correlation coefficient.

Implementing MSA_1 as a function in R (v4.4.3 [2025-02-28 ucrt], R Core Team, 2025) can be done with the following code:

```

1 # MSA #1 ('Mark II', Kaiser, 1970)
2 #
3 # @param R A correlation matrix
4 # @return A matrix of KMO indices, where columns are the individual KMO indices and
5 #         ↪ one overall KMO index
6 msal <- function(R = NULL){
7   if (!is.matrix(R)) stop("Input must be a matrix.")
8   if (!all(R == t(R))) stop("Matrix must be symmetric.")
9   inv_R <- solve(R)
10  P <- -inv_R / sqrt(outer(diag(inv_R), diag(inv_R)))
11  r2 <- R^2
12  p2 <- P^2
13  diag(r2) <- 0
14  diag(p2) <- 0
15  kmo_i <- 1 - (rowSums(p2) / rowSums(r2))
16  kmo_overall <- 1 - (sum(p2) / sum(r2))
17  return(list(overall = kmo_overall, individual = kmo_i))
18 }

```

6.3.2 Kaiser and Rice (1974)

The 'second' MSA (i.e., MSA_2), also known as 'Little Jiffy, Mark IV', by (Kaiser and Rice, 1974). The formula for calculating the overall MSA is provided in definition (7.4.2.1).

$$\text{MSA}_2 \equiv \frac{\sum_{i \neq j} \sum \hat{\rho}_{i,j}^2}{\sum_{i \neq j} \sum \hat{\rho}_{i,j}^2 + \sum_{i \neq j} \sum \hat{\rho}_{q[i,j]}^2} \quad (7.4.2.1)$$

where $\Sigma\Sigma$ denotes the row-wise summation operator, $\hat{\rho}$ denotes the estimated Pearson product-moment correlation coefficient, and $\hat{\rho}_q$ denotes the estimated anti-image product-moment correlation coefficient.

The formula for calculating individual MSAs are provided in definition (7.4.2.2).

$$\text{MSA}_2(i) \equiv \frac{\sum_{j,j \neq i} \hat{\rho}_{i,j}^2}{\sum_{j,j \neq i} \hat{\rho}_{i,j}^2 + \sum_{j,j \neq i} \hat{\rho}_{q[i,j]}^2} \quad (7.4.2.2)$$

where Σ denotes the summation operator, $\hat{\rho}$ denotes the estimated Pearson product-moment correlation coefficient, and $\hat{\rho}_q$ denotes the estimated anti-image product-moment correlation coefficient.

A function implementing MSA_2 in R (v4.4.3 [2025-02-28 ucrt], R Core Team, 2025) is provided in the code below:

```

1 # MSA #2 ('Mark IV', Kaiser & Rice, 1974)
2 #
3 # @param R A correlation matrix
4 # @return A matrix of KMO indices, where columns are the individual KMO indices and
5 #         ↪ one overall KMO index
6 msa2 <- function(R = NULL){
7   if (!is.matrix(R)) stop("Input must be a matrix.")
8   if (!all(R == t(R))) stop("Matrix must be symmetric.")
9   inv_R <- solve(R)
10  P <- -inv_R / sqrt(outer(diag(inv_R), diag(inv_R)))
11  r2 <- R^2
12  p2 <- P^2
13  diag(r2) <- 0
14  diag(p2) <- 0
15  kmo_i <- rowSums(r2) / (rowSums(r2) + rowSums(p2))
16  kmo_overall <- sum(r2) / (sum(r2) + sum(p2))
17  return(list(overall = kmo_overall, individual = kmo_i))
18 }

```

6.3.3 Kaiser (1981)

The 'third' MSA (i.e., MSA_3) proposed by (Kaiser, 1981), which can also be considered 'Mark V' of 'Little Jiffy'. The formula for calculating the overall MSA is provided in definition (7.4.3.1).

$$MSA_3 \equiv \sqrt{1 - \frac{\sum_{i \neq j} \sum \hat{\rho}_{q[i,j]}^2}{\sum_{i \neq j} \sum \hat{\rho}_{i,j}^2}} \quad (7.4.3.1)$$

where $\Sigma \Sigma$ denotes the row-wise summation operator, the *radical* (i.e., $\sqrt{}$) denotes the *square-root operation*, $\hat{\rho}_q$ denotes the estimated anti-image product-moment correlation coefficient, and $\hat{\rho}$ denotes the estimated Pearson product-moment correlation coefficient.

The formula for calculating individual MSAs are provided in definition (7.4.3.2).

$$MSA_3(i) \equiv \sqrt{1 - \frac{\sum_{j, j \neq i} \hat{\rho}_{q[i,j]}^2}{\sum_{j, j \neq i} \hat{\rho}_{i,j}^2}} \quad (7.4.3.2)$$

where Σ denotes the summation operator, the *radical* ($\sqrt{}$) denotes the *square-root operation*, $\hat{\rho}_q$ denotes the estimated anti-image product-moment correlation coefficient, and $\hat{\rho}$ denotes the estimated Pearson product-moment

correlation coefficient.

An R function implementing both definitions related to MSA_3 in R (v4.4.3 [2025-02-28 ucrt], R Core Team, 2025) is provided in the code below:

```

1 # MSA #3 ('Mark V', Kaiser, 1981)
2 #
3 # @param R A correlation matrix
4 # @return A matrix of KMO indices, where columns are the individual KMO indices and
5 #         ↪ one overall KMO index
6 msa3 <- function(R = NULL){
7   if (!is.matrix(R)) stop("Input must be a matrix.")
8   if (!all(R == t(R))) stop("Matrix must be symmetric.")
9   inv_R <- solve(R)
10  P <- -inv_R / sqrt(outer(diag(inv_R), diag(inv_R)))
11  r2 <- R^2
12  p2 <- P^2
13  diag(r2) <- 0
14  diag(p2) <- 0
15  kmo_i <- sqrt(1 - (rowSums(p2) / rowSums(r2)))
16  kmo_overall <- sqrt(1 - (sum(p2) / sum(r2)))
17  return(list(overall = kmo_overall, individual = kmo_i))
18 }

```

6.4 Characteristics of the KMO index

The results from the Monte Carlo simulation study of the characteristics of the KMO index discussed in section 2.3 are provided here. Results from the first set of simulations investigating the relationship between the inter-correlations and the anti-image inter-correlations are illustrated in figure 3 in section 6.4.1. Results from the second set of simulations used to gauge the relationship between the overall KMO index and linearly independent inter-correlations are shown in figure 4 in section 6.4.2. Results from the third and last set of simulations that sought to assess the relationship between the overall KMO index and linearly interdependent inter-correlations are shown in figure 5 in section 6.4.3.

6.4.1 Association between correlations & anti-image correlations

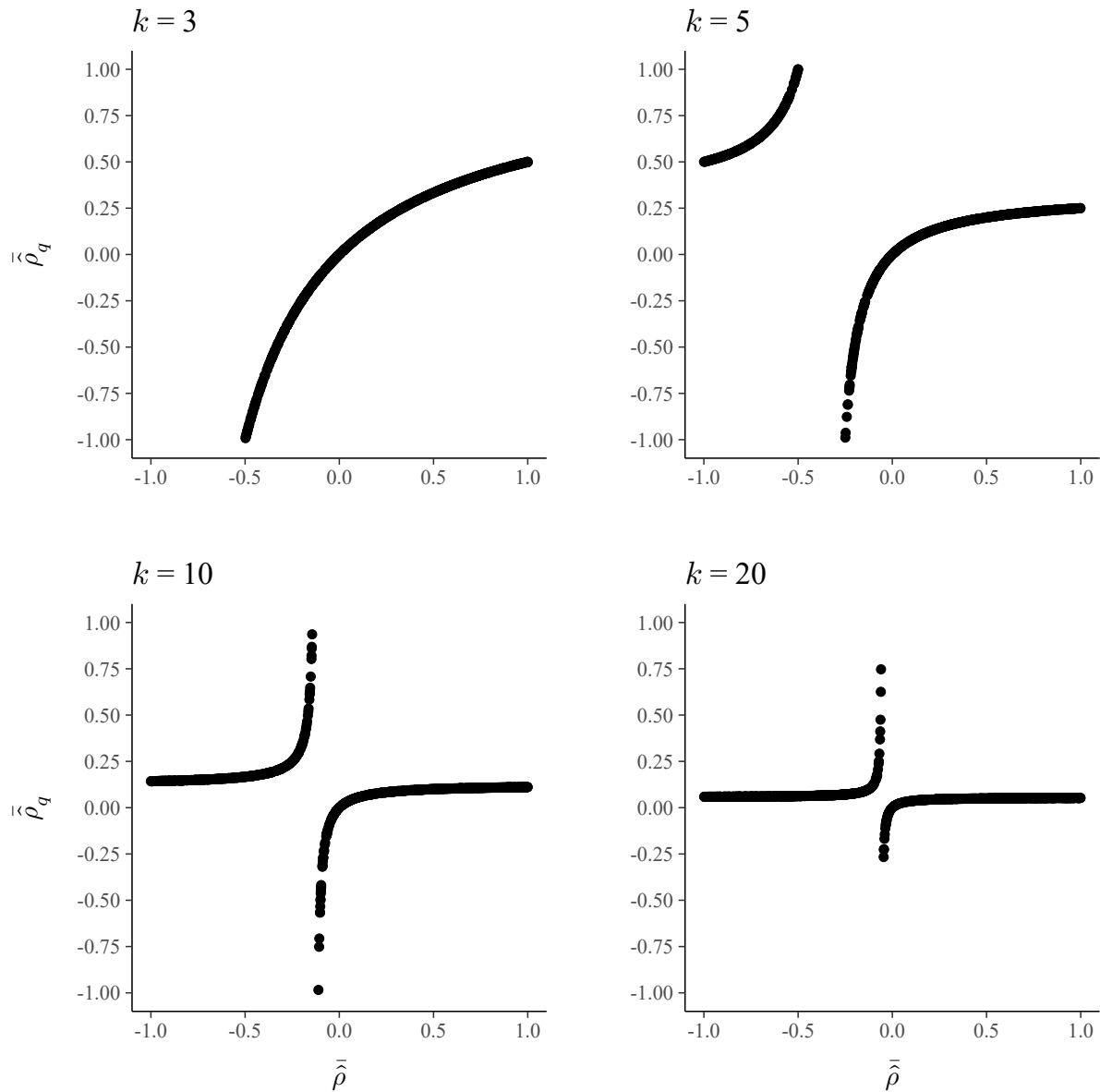


Figure 3. Association between correlations & anti-image correlations

NOTE: $\bar{\rho}$ is the mean estimated inter-correlation; $\bar{\rho}_q$ is the mean estimated anti-image inter-correlation; k is the number of measured manifestations. Metrics computed using frequentist formula (see Section 2.2). Simulated data, with 1,000 of 40,005 simulations randomly sampled for display in each panel.

6.4.2 Independent correlations

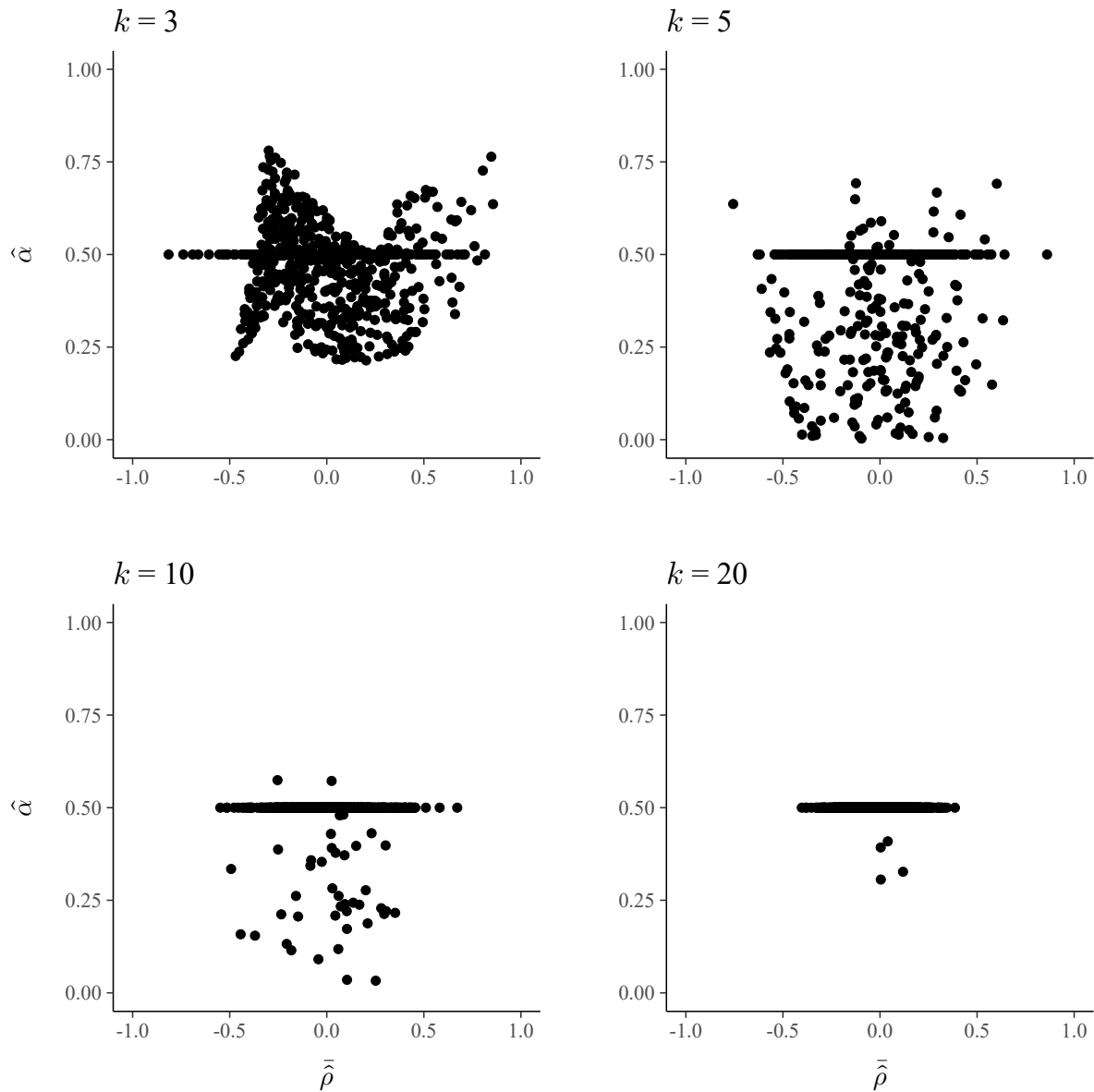


Figure 4. KMO index characteristics with independent correlations

NOTE: $\bar{\rho}$ is the mean estimated inter-correlation; $\hat{\alpha}$ is the estimated overall 'sampling adequacy' (KMO index); k is the number of measured manifestations. Metrics computed using frequentist formula (see Section 2.2). Simulated data, with 1,000 of 40,005 simulations randomly sampled for display in each panel.

6.4.3 Interdependent correlations

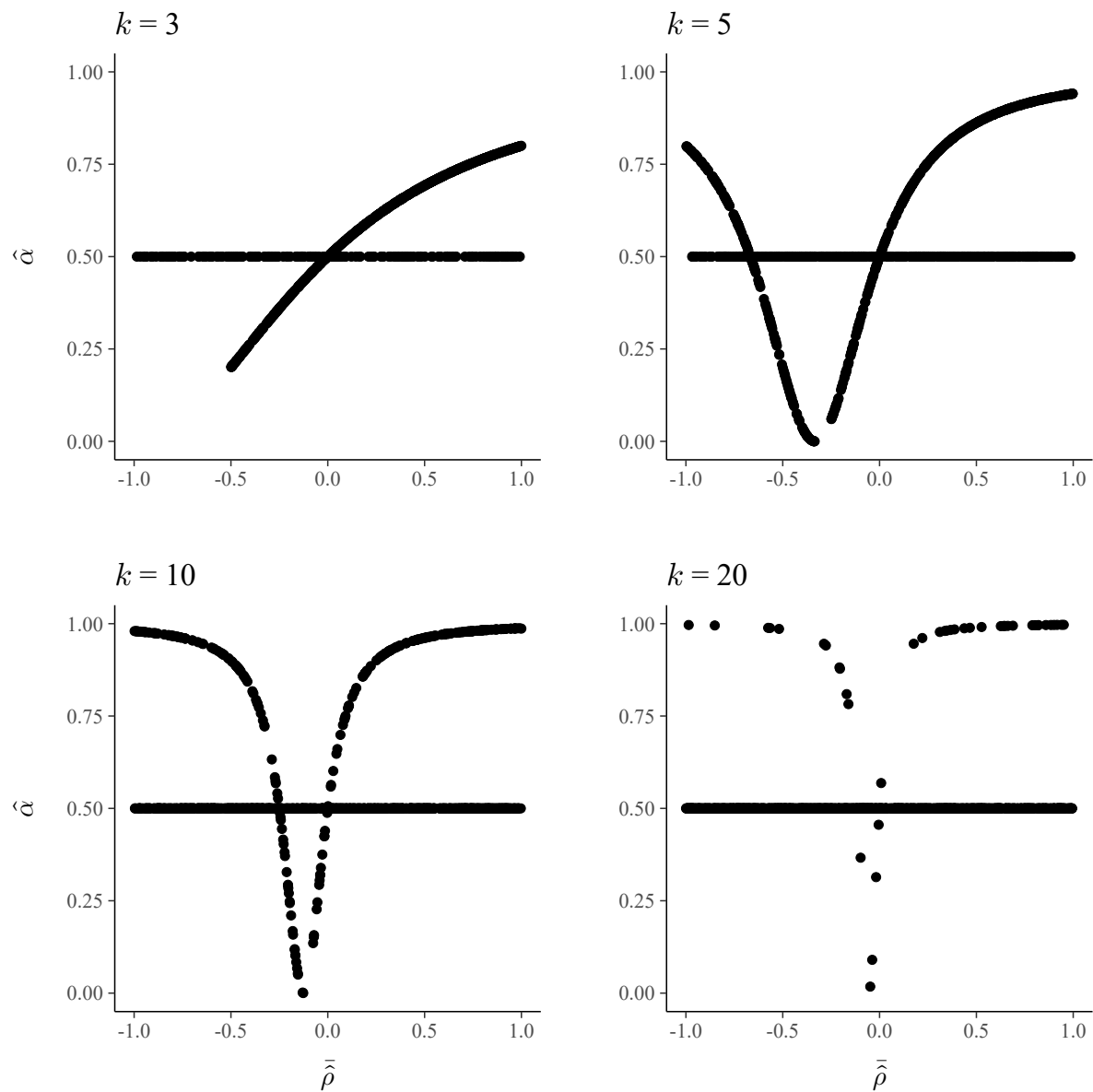


Figure 5. KMO index characteristics with interdependent correlations

NOTE: $\bar{\rho}$ is the mean estimated inter-correlation; $\hat{\alpha}$ is the estimated overall 'sampling adequacy' (KMO index); k is the number of measured manifestations. Metrics computed using frequentist formula (see Section 2.2). Simulated data, with 1,000 of 40,005 simulations randomly sampled for display in each panel.

6.5 Results matrix

The matrix of results from the Monte Carlo simulated data covered in section 2.4 is provided here. The lower triangle of the matrix in table 2 showcases the estimated inter-correlations (i.e., $\hat{\rho}$), with the upper triangle containing the estimated anti-image inter-correlations (i.e., $\hat{\rho}_q$), while the diagonal displays the estimated individual KMO indices (i.e., $\hat{\alpha}_m$), one for each manifestation (θ).

Table 2. Results matrix

	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8	θ_9	θ_{10}
θ_1	0.844	0.016	0.066	0.125	0.053	-0.003	0.040	0.051	0.081	0.005
θ_2	0.101	0.852	0.053	0.156	-0.011	0.072	0.040	0.036	0.081	0.075
θ_3	0.130	0.139	0.869	0.046	0.010	0.076	0.063	0.078	0.048	0.079
θ_4	0.222	0.277	0.186	0.820	0.044	0.131	0.108	0.125	0.177	0.131
θ_5	0.115	0.080	0.094	0.176	0.836	0.073	0.111	0.090	0.014	0.107
θ_6	0.101	0.187	0.174	0.288	0.169	0.857	0.108	0.054	0.078	0.103
θ_7	0.141	0.163	0.166	0.280	0.200	0.239	0.851	0.070	0.147	0.054
θ_8	0.136	0.142	0.164	0.260	0.172	0.179	0.195	0.864	0.048	0.064
θ_9	0.179	0.210	0.167	0.342	0.138	0.232	0.281	0.189	0.840	0.121
θ_{10}	0.113	0.194	0.180	0.296	0.196	0.239	0.210	0.192	0.264	0.851

NOTE: Estimated inter-correlations ($\hat{\rho}$) provided in the lower triangle, estimated anti-image inter-correlations ($\hat{\rho}_q$) provided in the upper triangle, and estimated individual KMO indices provided in the diagonal. Based on simulated data ($n = 1,000$) and computed using frequentist estimators (see Section 2.2).